



PIPE SUPPORTS

PURPOSE

This document establishes guidelines and recommended procedures for the design of pipe and electrical supports. It has been developed to establish a single point of reference for underlying design philosophies and theories pertaining to the design of structures which primarily support piping or electrical loads, including those which support equipment loads such as air coolers.

SCOPE

This guideline contains the following major sections:

- General - An overview of pipe supports
- Design Loads - A discussion on types of loads resisted by pipe supports and their origins
- Design of Pipe Support Components - A more detailed discussion on various components of pipe supports
- Pipe Bridges - A discussion on design considerations for components of pipe bridges
- Miscellaneous Pipe and Electrical Supports - A discussion on design considerations for various types of miscellaneous supports
- Revamp of Existing Supports – References that address considerations for revamp of pipe supports
- Cost Competitive Engineering and Design – A discussion and references that address considerations for cost competitive engineering and design of pipe supports

APPLICATION

This guideline should be used by the Lead Structural Engineer when developing project design criteria and by Structural Engineers during the design of structures supporting loads from piping or electrical components, including equipment loads such as air coolers.

Theories and design procedures herein may be applied to the design of components of other structures, such as equipment structures and substations, which support piping or electrical loads.

This guideline may be used as a reference when designing pipe rack modules, however details specific to modular construction and unique loads associated with module transportation and erection should be based on the best practices associated with modular construction.

Project specific design requirements can vary widely from project to project based on client preferences, local codes, availability of material or labor, geographic design considerations and many other factors. Due to this, it is not possible to address every potential detail in this guideline. Rather, this guideline covers the fundamental design



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considerations of pipe supports, relying on engineering knowledge and judgment to account for design requirements beyond the scope of this document.

GENERAL

The term "pipe supports" describes a class of concrete or steel structures ranging from simple supports carrying light utility piping or single pipes to major, multilevel frames supporting process and utility piping plus air coolers and other equipment at top.

Pipe supports include the following types:

- Strutted concrete or steel pipe racks
- Unstrutted concrete or steel secondary pipe racks
- Pipe bridges and similar structures intended to span longer distances between columns to grade.
- Sleeper supports
- Miscellaneous Pipe and Electrical Supports (MPS or MES)
 - Single column supports
 - Two column supports
 - Cantilevered supports
 - Trapeze type hangers
 - Field supports provided by Piping or Electrical Engineering which often require verification by Structural Engineering
 - Any other structure intended to support pipe or electrical loads

Pipe Racks

The term "pipe rack" is used to describe a specific subset of pipe supports. Pipe racks are multi-level frames, strutted together to form a structure that serves as the primary means of supporting pipe within process units, sometimes referred to as "process unit" pipe racks. Pipe racks supporting pipe between units are known as "interconnecting" or "outside battery limits" pipe racks and usually allow the Structural Engineer greater flexibility with the configuration of bracing, struts and frame design. Secondary pipe racks, also known as "finger" or "lateral" pipe racks, are often used within process units when it is necessary to support piping or electrical components from the main unit pipe rack outward toward the edge of the unit. It is not uncommon for main pipe racks to support air coolers or other equipment.

Note: *Pipe racks are also referred to as "pipeways" and sometimes as one word, "piperacks".*



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Geometry and Spacing

The spacing of supports is based on the allowable span for pipe and electrical cable tray being supported and is coordinated with Piping and Electrical Engineering. Typical support spacing is 20 to 25 feet (6.0 m to 7.5 m). However, support spacing can vary from 10 to 40 feet (3.0 m to 12.0 m) utilizing intermediate beams to support items not capable of spanning the entire length. Pipe bridges are typically used when the spacing between support bents exceeds 40 feet (12.0 m), which normally occurs at road crossings.

The transverse dimension of the support is determined based on a combination of space and separation requirements of the items being supported and the number of levels the support will be. When a pipe rack is required to support air coolers, the width of the support will usually be determined based on the geometry of the equipment being supported. Typical transverse frame dimensions for a pipe rack supporting air coolers is 20 ft to 30 ft (6.0 m to 8.0 m). Clearances over, under and around pipe supports are an important consideration in design.

Note: *Longitudinal refers to the direction along the axis of the items being supported and transverse refers to the direction perpendicular to the items being supported. These are illustrated in Attachment 03.*

Strutted and Unstrutted Supports

Pipe supports are usually designed as cantilevers or rigid frames (pipe bents) in the transverse direction. However, when transverse bracing can be used, significant advantages, such as cost savings, can be realized. In the longitudinal direction, supports can be unstrutted and designed as cantilevered columns, or strutted together in order to transfer longitudinal loads to a common set of vertical bracing, or less commonly to a rigid frame. The design approach used for each direction will depend on the project design criteria.

Note: *When utilizing a rigid frame in the longitudinal direction, it is more common to use concrete than steel due to the increased weak axis strength of rectangular concrete sections.*

Material of Construction

The material of construction should be established in the FEED phase of a project and is dependent on several factors such as:

- Site conditions and accessibility
- Fireproofing requirements (it is sometimes more economical to use concrete pipe supports in lieu of fireproofed steel supports)
- Procurement considerations such as material availability or delivery times
- Modularization



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- Client preferences
- Anticipated modifications during construction or future work
- Labor and other costs associated with construction

It is important to consider the engineering schedule when making decisions about materials of construction. Concrete supports can allow for longer home office engineering time due to shorter lead times required to fabricate rebar and pre-cast concrete versus the procurement cycle of developing steel detail drawings, reviewing the drawings, fabrication, and delivery of structural steel to the jobsite. However, concrete supports are generally more difficult to alter after fabrication and therefore require more planning and project development before the design can be issued for construction.

Fireproofing

Requirements for fireproofing are established by client requirements, industry standards and local regulatory requirements. Fireproofing will usually be required on supports within process units where they would be subject to fire from equipment and other potential sources. However, fireproofing may not be required on supports outside the limits of the process units and other areas where fire is not a concern. Spring hangers and other devices where fireproofing would prevent normal operation, as well as minor supports such as hangers and “dummy legs”, typically are not fireproofed. In these cases, a fireproofed “catch beam” or other means of protection from failure during a fire is usually provided below the pipe in order to support the pipe with minimal deflection.

Concrete is the most commonly used material for fireproofing. Other proprietary light-weight material, including cementitious and intumescent coatings, can be used when the additional costs of the material is offset by the advantages of using the lighter weight material, as in the case of high seismic demand or modular construction.

Typical fireproofing requirements for pipe supports will specify a minimum height from grade (or other level capable of supporting pool fires) and whether fireproofing is required up to the level of support for air coolers or other equipment. As is the case with other structures, when fireproofing is required, all members required to prevent immediate collapse of the pipe support and further escalation of the fire, such as knee bracing supporting cantilevers or struts supporting piping, should be fireproofed. Steel connection details must consider blockouts or other special detailing needs in order to erect the structure in the most cost effective manner.

Guideline 000.215.0760 and Standard Drawings 000.215.4060 and 000.215.4061 contains more specific information on defining the extents of fireproofing and fireproofing details.



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DESIGN LOADS

Gravity Loads

Gravity loads come from many sources including:

- Self-weight of the structure
- Fireproofing
- Weight of pipe, electrical components & equipment
- Operating loads
- Platform live loads
- Snow and ice

Self-Weight of Structure

The weight of all structural members and connections, including fireproofing, should be considered in the design of the pipe support.

Self weight of fireproofing should be based on all factors affecting the weight including the density of material being used, thickness required per project requirements and whether blocked or contoured fireproofing is used.

Piping Gravity Loads

The piping gravity load to be used in design should be agreed to at the start of engineering and documented in the project design criteria. Historically, 40 psf (2.0 kN/m²) has been used as a uniformly distributed load (UDL) to account for the dead load of pipe. (This corresponds to NPS 8 (DN 200) STD pipe full of water spaced at 15" (375 mm).) However, assuming 40 psf (2.0 kN/m²) is considered conservative since actual piping loads on process unit pipe racks are typically closer to an equivalent UDL of 25 psf (1.2 kN/m²). Conversely, heavily loaded levels can have an equivalent UDL can be as high as 60 psf (3.0 kN/m²). Due to this wide range of loads, it is good practice to coordinate with Piping Engineering to determine the appropriate loads to use for design, challenging client criteria and other sources of loading criteria when necessary. A properly determined UDL will be one which requires consideration of one or two point loads for heavier pipes, as described below, than accounted for in the UDL. Additional information for determining proper UDL's can be found in Attachment 01.

When pipes of a larger diameter or thicker pipe wall thicknesses are used, it is necessary to add a point load to the supporting member to compensate for the additional dead load not accounted for in the UDL. This generally corresponds to a NPS 12 (DN300) STD pipe or a NPS 10 (DN250) XS pipe when a UDL of 25 psf (1.2 kN/m²) is being used.

$$P = s(w - qd), \text{ where}$$



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P = Additional concentrated load kips (kN)

s = Support spacing ft (m)

w = Weight of pipe per unit length kip/ft (kN/m)

q = UDL kip/ft² (kN/m²)

d = Pipe diameter ft (m) + 2 x Insulation thickness ft (m) + clear space to adjacent pipe (not to exceed 8" (200mm)), or a credit for adjacent pipes significantly smaller than the average pipe size.

The above listed loading may be used individually on different levels in a pipe support based on the availability and type of layouts.

The empty load contribution, historically, is assumed to be 60% of the operating piping load.

The normal operating weight of vapor and flare pipes, pipes with commodities significantly lighter than water, and NPS 20 (DN500) and larger pipes should be established early. They are likely to have a large enough difference to affect the design of the pipe support, especially for load combinations including wind or seismic loads.

Electrical Loads

Several variables influence the actual load of electrical components to be used when designing electrical support elements including:

- Strength of cable tray
- Size of cables required
- Spacing of supports (shorter spans allow larger loads)
- Expected snow and other non-dead load the cable trays must support
- Spacing of cable trays on the support

Due to the wide range of loads and the factors listed above, Electrical Engineering should be consulted to determine the design weight, layout and support requirements of electrical cable tray and conduits. UDL's usually fall within the range of 10 psf to 40 psf (0.5 kN/m² to 2 kN/m²).

The Structural Engineer also needs to work with Electrical Engineering to understand the support requirements of electrical cable tray and conduits. Electrical components generally require more supports than piping does, especially at changes in direction and vertical runs.

When performing preliminary design for estimates or when no other information is available, an estimated UDL of 20 psf (1 kN/m²) can be used to design elements



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supporting a single level of cable tray. The estimated 20 psf (1 kN/m²) load covers the weight of the cables and the cable tray itself, considering required clear spacing between cable trays. It does not include the effects of snow, ice and other atmospheric loads which may accumulate within the side rails of cable trays.

Note: *The maximum allowable cable tray loads published in manufacturer's catalogs includes the effect of all loads the cable tray would be subjected to, including snow and ice. Engineering judgment and current best practices should be used to determine and apply snow, ice and other loads.*

Equipment Loads

Weights for air coolers and other supported equipment, including associated platforms, walkways and other components provided as part of the equipment package, should be obtained from supplier's engineering data. When estimated loads are needed, Mechanical Engineering should be consulted.

Note: *Due to the wide range of variables associated with determining approximate weights of equipment, such as number of tube bundles within an air cooler, size and number of fans required, weight of operating liquids, and other process specific requirements, it is not practical to propose estimated equipment weights in this document. Mechanical Engineering should be consulted for all equipment weights estimates.*

Test Load

The test load of pipes and other items to be hydrotested can be simply calculated by adding the empty weight of the component to be tested to the weight of the test medium, typically water, necessary to fill the volume. Refer to Attachment 01 for the weight of water required to fill pipes. If a medium other than water is being used, it up to the responsible Piping or Mechanical Engineer to identify those items and communicate the anticipated weight to the Structural Engineer.

Usually, the UDL determined in the "Piping Gravity Load" section will result in sufficient strength to support piping during the hydrotest phase. However, the Structural Engineer should exercise due diligence to identify large diameter pipes, especially vapor and flare lines with significant differences between operating and hydrotest weights, and other items which when filled with the test medium would warrant additional point loads to be included in load combinations considering test loads. Hydrotest point loads generally influence local design of members, and will not govern the overall design of the pipe support except for smaller supports. It is unlikely that all pipes will be tested simultaneously, and it is generally not required to design for this condition.



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Snow and Ice Loads

Pipe supports should be designed considering the effects of snow, ice and other weather related loads. The effects of weather related loads can have a direct impact to the structure, such as increased vertical loads, or a secondary impact, such as increased wind profiles due to ice accumulations. In all cases, the loads should be determined in accordance with the project design criteria based on applicable codes.

The total snow load on the structure is 80% of the ground snow load, similar to a flat-roofed, heated building in design codes. For distribution of snow loads on pipe racks assume 50% of total snow load accumulates on the top level and the balance is distributed equally in levels below. For pipe racks with air coolers or similar components that would prevent the distribution of snow loads to lower levels, snow load shall be considered on the top level only. The applicability of the previous assumption should be verified by the Structural Engineer, based on experience, prior to implementing in the design of the pipe support. The Structural Engineer should also consider adjacent structures which could cause drifting on the pipe rack, thereby increasing the design snow load.

In addition to the assumption presented above, application of snow and ice loads should be in accordance with applicable codes and industry best practices; including application to structural members, large diameter pipes and other items capable of collecting snow or ice loads which should be considered in the design of the pipe support.

Less Common Loads

In addition to the major gravity loads contribution from sources previously discussed, many other potential sources of gravity loads can contribute to the design of pipe supports.

- Large valves and valve actuators, weighing more than 2 kip (9 kN)
- Complex piping such as manifolds and very high pressure pipes
- Operating conditions which change the load distribution of supported elements such as thermal growth lifting pipes off of supports
- Complex electrical configurations

The Structural Engineer should carefully consider these and other sources of potential gravity loads to ensure the support is adequately designed to carry the intended loads.

Lateral Loads

The lateral loads discussed below include wind, earthquake, friction and anchor loads, including guide loads. These loads are described as acting in the transverse direction or the longitudinal direction.



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Wind and Earthquake Loads

Wind loads should be applied according to Guideline 000.215.1215 Wind Load Calculation.

Earthquake loads should be applied according to Guideline 000.215.1216 Earthquake Engineering.

For long pipe racks, the longitudinal design is usually governed by seismic, friction or thermal anchor loads due to the minimal profile exposed to wind in the longitudinal direction. However, consideration should be given to components of the pipe rack which might be exposed to the full longitudinal wind loads in order to design the localized load path for the most severe loading.

Consideration should be given to the friction resistance offered by piping systems to resist wind loads.

Friction Loads

Friction loads are caused by the thermal growth or contraction of pipes as they slide on the pipe support during startup, shutdown and operation. Due to the uneven thermal strain in pipes at any given time, friction forces of individual pipes are assumed to be partially resisted by adjacent pipes. In order to provide a nominal unbalanced friction load acting on the pipe support, a resultant friction load equal to a percentage of the total pipe weight tributary to that pipe support is assumed depending on the number of pipes on a pipe support as follows.

- 10% of the total pipe weight for number of pipes > 6
- 20% of the total pipe weight for number of pipes = 4, 5, or 6
- 30% of the total pipe weight for number of pipes = 2, or 3
- 40% of the total pipe weight for number of pipes = 1

Consideration should be given to circumstances where heavier pipes acting alone produce a larger friction load than the group of pipes associated with the heavy pipe, as would occur when considering 40% of the heavy pipe versus 10% of UDL on a pipe rack. These different friction loads could lead to different governing factors such as one governing weak axis bending, and another weak axis shear.

Note: *The 40% coefficient of friction represents galvanized steel to galvanized steel with an allowance for fouling. When different materials of construction are in contact, the factors above should be modified accordingly.*

Loading conditions could arise where the reduced percentage for friction loads for a group of pipes might be unconservative and should not be used, such as near expansion loops when all pipes are likely to be moving in the same direction at the same time. The



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Structural Engineer should carefully consider such circumstances and apply the appropriate friction factors when designing the support.

Friction loads should be treated with the same level of certainty as dead loads for strength design, unless there is sufficient cause to use a higher load factor when justified by uncertainty. However, in load combinations considering reduced dead loads, caution should be used to ensure loads causing instabilities or higher stresses compared to gravity loads are not lowered using the reduced dead load factors.

The friction loads prescribed above are to be used for local member design only. Unless a higher load is justified, 50% of the prescribed local friction design loads should be considered for global design of the pipe support system. Further information can be found in 000.215.STC01015.

Anchor Loads

Overview

Piping anchors (and guides) cause expansion movement to react at desired locations in a piping system. The Pipe Stress Engineer is responsible for locating anchors and providing anchor loads. For this reason, it is important that the Structural Engineer communicate with the Pipe Stress Engineer prior to starting and during the design of any pipe support.

The loading direction of anchor load should be determined and given by the Pipe Stress Engineer with clear indications of how many loads will act in same direction or cancel each other out during normal operation.

Anchor loads which are small are usually resisted by frictional resistance from adjacent pipes. Special consideration should be given to pipe supports on which all or most of the pipes are anchored, or on which significant anchor loads are anticipated. Ideally, the Pipe Stress Engineer will anchor one or two pipes on a single support in order to evenly distribute the anchor loads throughout the length of the pipe supports, or to anchor all of the pipes at a single support, as close as possible to the lateral force resisting frames, in order to localize the effects of the anchor loads.

Anchor loads should be treated with the same level of certainty as dead loads for strength design, unless there is sufficient cause to use a higher load factor when justified by uncertainty. In such cases, higher load factors should be applied only to the specific component of the anchor load subject to the higher uncertainty. However, in load combinations considering reduced dead loads, caution should be used to ensure loads causing instabilities or higher stresses compared to gravity loads are not lowered using the reduced dead load factors.

Note: *The Structural Engineer should consult with the Pipe Stress Engineer to verify what, if any, load factors or allowances for uncertainty have been*



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included in the derivation of the provided loads and adjust the structural design load factors accordingly.

Components of Anchor Loads

The most common components of anchor loads are thermal loads, friction loads, wind loads and seismic loads. Other less common contributors, such as slug loads, thrust loads and impact loads, may be defined for specific projects and should be reviewed with the Pipe Stress Engineer to understand the origin of the loads, the transient and steady state characteristics of the loads and any other variables to be considered when applying the loads to the supports using appropriate load factors and load combinations.

Thermal

The thermal components are those loads caused by restricting thermal growth between two support points in a pipe or the amount of load generated when pipes are allowed to expand creating a spring load on perpendicular runs of pipe, restrained against transverse movement. Thermal anchor loads can be in the vertical or horizontal direction.

Unrestrained thermal growth results in friction loads and anchors or in a redistribution of dead load if pipes are allowed to lift off of supports.

Friction

The friction components are those loads that resist the sliding of pipes on supporting beams. Friction anchor loads represent the potential unbalanced friction forces from opposite sides of an anchor location.

Wind

The wind components are those loads caused by wind pressures on the supported piping system.

Seismic

The seismic components are those loads caused by the inertial mass effects of the supported piping system in response to earthquake ground motions.

Global vs. Local Design

Separate considerations should be given to local effects and global effects on the structure when designing for anchor loads.



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The full magnitude of the defined anchor loads should be used to locally design the supporting beam, connections and other components of the support until the load is sufficiently distributed into the global system, usually a column or column line.

Global systems are usually designed for a reduced percentage of the defined anchor loads based on several considerations by the Lead Structural Engineer, such as:

- Size of the overall pipe support system
- Total number of anchor loads applied to the system
- Total magnitude of anchor loads applied to the system
- Likelihood of all anchors acting in the same direction at the same time

Note: *Wind and seismic components of anchor loads are typically applicable to the local design of members. Global wind and seismic loads are to be determined in accordance with 000.215.1215 and 000.215.1216.*

Early Design without Actual Anchor Loads

Since anchor loads are normally not available until the latter stages of a project, steel pipe supports are often designed without considering anchor loads. As the anchor loads become available, supports are then checked individually for the actual anchor load and reinforced if necessary. This generally leads to the lowest cost alternative, particularly if modifications can be made in the steel fabricator's shop or at site before erection.

In cases where modifications to the pipe supports are costly or not easily performed, assumed anchor loads should be considered in the design when actual anchor loads are not available. These assumed loads should be developed with the Pipe Stress Engineer and should include considerations for application of the loads for local and global design, and to identify specific pipes which are likely to exceed the nominal anchor load being assumed. Assumed anchor loads should be on the order of magnitude of 2 kips (9 kN), applied at mid span of the beam. The assumed load should be considered as 30 percent thermal component, and 70 percent the friction component.

Other Loads

Stresses produced by the variation in the atmospheric temperature should be considered in the engineering and detailing of pipe supports. The differential temperatures should be defined in the project design criteria. For a cost competitive solution, restraint against expansion and contraction should be minimized with considerations such as:

- Slide plates for heavy loads
- Isolation of braced bays by eliminating struts or expansion joints utilizing slotted holes
- Use a single level of continuous struts when not precluded by wind or seismic requirements



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Application of other loads such as Pressure Safety Valve (PSV) loads, surge loads and slug loads should be per Piping or Process Engineering, or as stated in vendor information.

Load Combinations

Load combinations used in the design of pipe supports should reflect those defined in the project design criteria. The load combinations as outlined in this guideline are indicative of most critical load combinations usually expected. Pipe supports will normally be designed to resist the following combinations of loads.

- Gravity loads (empty, operating, or test conditions).
- Gravity loads (empty) + transverse wind loads or earthquake loads.
- Gravity loads (operating) + transverse wind loads or earthquake loads + thermal component of anchor loads and guide loads.
- Gravity loads (operating) + friction loads + thermal component of anchor loads and guide loads.
- Gravity loads (operating) + anchor loads (friction and thermal components) + guide loads.

Note: *Facilities in hurricane regions will typically shut down when forecasted winds approach design strength. In hurricane regions, the load combinations used should reflect a combination of reduced design wind loads with the operating conditions described above, and the full design wind loads with appropriate empty conditions resulting from a closed plant. These considerations should be discussed with and agreed to by the Client.*

It is normally assumed that friction loads, including the friction component of anchor loads, do not occur at the same time as wind or earthquake loads.

Project design criteria requirements sometimes stipulate that test loads be combined with a reduced wind load.

When determining dead loads and other stabilizing loads, it is important to consider the actual loading on the supports. Circumstances could exist where assuming the full UDL is un-conservative.

- When piping or electrical loads are significantly lighter or not present on one side of a pipe support
- When the operating load is not expected to occur at the same time as design strength lateral loads such as hurricane wind loads

This is not a complete list of possibilities and engineering judgment should be exercised in defining the appropriate resisting loads for uplift or load reversal cases.



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Likewise, consideration should be given to ensure loads which cause instability are applied in the direction which results in the highest overturning and uplift. However, not all designs will require the maximum instability to be compared to the minimum resistance. Engineering judgment should be used to ensure the combinations being used are logical and likely to occur at the same time.

DESIGN OF PIPE SUPPORT COMPONENTS

The design of pipe support components is generally governed by stress. At times however, deflections and settlement of pipe supports merit special consideration as discussed below.

Refer to Attachment 03 for figures depicting the components of pipe supports discussed below including

- Rigid frames (bents)
- Longitudinal struts
- Vertical bracing
- Connections
- Foundations
- Pipe bridges
- Miscellaneous supports

Method of Analysis

The decision to perform a manual design, or a computer design either in 2D or 3D should be based on the degree of complexity of the pipe support and the accuracy of results desired. Simple cantilever pipe supports are most efficiently and easily designed using manual design methods in lieu of computer analysis. Simple pipe racks should be designed using two separate 2D computer analysis models in lieu of 3D computer analysis. Only the most complicated pipe racks should be analyzed using 3D analysis, and should be in agreement with the Lead Structural Engineer.

Consideration of the envelope of column base stiffness conditions is important to the development of a cost-effective design. The envelope of conditions regarding the actual stiffness would include the load response of the foundation elements, soil, base plate, and anchor bolts. The extreme limits of the range of rotational stiffness are the pinned and fixed conditions. The extreme limits of the range of lateral stiffness are the roller and fixed conditions. Some aspects of the design will be governed by more rigid potential base behavior and other aspects will be governed by more flexible potential base behavior. Therefore, a single analysis run may not be a suitable basis for all aspects to be designed.



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Deflection of Pipe Supports

The deflection of structural members in a pipe support may be the governing consideration in the design of the piping system. Whether the deflection being considered is a transverse beam due to pipe anchors or lateral deflection of the frame due to wind or seismic loads, the criteria for deflection should be specified in the project design criteria.

Where the Pipe Stress Engineer indicates that the lateral deflection of a particular directional anchor is critical, where multiple anchors are located on the same frame, or where there is an absence of other pipes to provide restraint, the deflection should be calculated and reviewed with the Pipe Stress Engineer for concurrence.

Rigid Frames

A structural analysis of a transverse frame composed of transverse beams and columns should be performed to determine all forces, reactions, and displacements produced by the loads and load combinations given above.

Precast concrete frames, and steel frames intended to be assembled prior to erection, should be analyzed for handling stresses induced from being transported and lifted. The location and details of lifting points should be indicated in the drawings.

Transverse Beams

Transverse beams should be designed to resist all forces and moments calculated from the structural analysis.

For the flexural design of steel beams, the unbraced length of the top flange can be considered one third of the span or less when there is sufficient pipe load to provide stability to the top flange. The unbraced length of the bottom flange should be considered to be the full length of the beam unless stability is provided by stiffeners, braces or other means of stability.

In general, torsion resulting from horizontal movement of pipes across the top flange of the support beam is assumed to be resisted by the counteracting forces of adjacent pipes on the same support. However, torsion should be considered when significant loads, such as large anchor loads, are applied to the beam flange. When torsion is a design consideration the torsion should be considered using engineering principles found in sources such as Salmon and Johnson. Torsion may be reduced using horizontal bracing to redistribute the loads causing the torsion or by providing additional longitudinal beams capable of resisting the torsion. Refer to Attachment 02 for a more detailed discussion of torsion.

Intermediate transverse beams are sometimes required to reduce the span for smaller pipes and cable trays and may be required in pipe bridges. The loading on the



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intermediate beams should be based only on piping and cable trays supported on these beams.

Columns

The columns must be capable of resisting all axial, moment and shear loads calculated from the structural analysis. The analysis should be made using the following column base conditions.

- Steel pipe supports

Strutted - fixed base in both the transverse and longitudinal directions, or pinned base in both the transverse and longitudinal directions, with the major axis of the column in the transverse direction. In general, the fixed base condition results in a smaller superstructure and a larger foundation with smaller lateral deflections. The pinned base condition results in a larger superstructure and smaller foundations with larger lateral deflections.

Unstrutted - fixed bases in both directions, with the major axis of the column in the longitudinal direction. A common design concept is to provide bracing in the transverse direction.

- Concrete pipe supports

Fixed at the foundation.

Longitudinal Struts

Longitudinal struts are used to transmit lateral loads to the lateral load resisting system, and to support gravity loads of piping and electrical items entering or exiting the pipe racks. They should be designed as beam-columns and meet all the code requirements for collector elements. Struts also serve to reduce the effective length of the pipe support columns.

Unless otherwise specified in the Project Design Criteria, where struts will support piping or electrical loads entering or exiting pipe racks, struts should be designed for the greatest of 50 percent of the gravity loading on the most heavily loaded transverse beam or the actual loading. The 50 percent loading accounts for the usual electrical and piping takeoffs. This loading should be excluded from the global design combinations unless there is a defined impact on the design of the columns, foundations or other elements of the pipe support. Prior to issuing any pipe support drawings AFC (Approved for Construction), the Structural Engineer should check piping layouts to verify that any struts subjected to unusually large loads have been given special consideration.



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Struts should be designed to resist axial forces produced by longitudinal loads. For normal conditions, longitudinal loads may be assumed to be transmitted to the struts at each column without reconsidering column bending in combination with the rigid frame analysis. However, if the vertical dimension between transverse beam and the strut in question is large (exceeding 3 feet (1.0m)), or large anchor loads occur on the transverse beam, the column stresses must be reconsidered.

Vertical Bracing

Vertical bracing is used to transmit longitudinal loads from the struts to the foundations. Braced bays should be spaced considering the magnitude of lateral loads, seismic detailing requirements, thermal isolation and constructability issues. Although there is no upper bound for braced bay spacing, maximum practical distance of a pipe support from a braced bay is around 100 ft (30 m).

Operating access is an important consideration when locating bracing. The Structural Engineer should coordinate the placement of bracing with Piping and Electrical Engineering. Vertical bracing in the transverse direction, when approved, can significantly reduce the amount of steel required in the transverse pipe support frame.

Horizontal Bracing

Horizontal bracing is not normally incorporated into the design of pipe supports. However, when conditions exist which require the use of horizontal bracing, such as horizontal trusses of pipe bridges or transfer of large anchor loads, careful coordination is required with Piping Engineering in order to avoid conflicts with vertical piping.

Connections

Connections should be designed in accordance with the applicable codes, project design criteria and the project standard drawings. Where the standard details are not appropriate or adequate, proper details need to be shown on the construction drawings. Connection selection and detailing should consider factors such as space allocation for piping and electrical components, load transfer, constructability, fabrication and seismic detailing requirements.

The guidelines listed below contain design information for connections and anchorage commonly used in pipe supports.

- Guideline 000.215.1201: Single Plate Connections
- Guideline 000.215.1203: Bracing Connections
- Guideline 000.215.1208: Base Plate Design Criteria
- Guideline 000.215.1209: Bolted End Plate Moment Connections
- Guideline 000.215.9885 Steel Connections – Details and Relative Cost



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- Guideline 000.215.STE05121: Anchor Bolt Design Guide

Other Connections

For connections between steel struts and concrete columns embedded items cast into the column or through bolts with sleeves cast through the bents may be used. Post-installed anchors should be avoided due to potential interference with reinforcing steel during installation. Fabrication and erection tolerances should be considered when specifying details in order to allow for fit up. Slotted holes and shim plates are two possible means to provide fit up allowances.

Foundations

Foundation design parameters are normally specified in the project design criteria or the geotechnical report. The type of foundations to be used should be compatible with the site conditions. Foundations should be designed using the support reactions for each direction at the column bases. When appropriate, the foundation may be incorporated in the analysis to reasonably represent the structure-foundation behavior. However, in all cases, the rigidity of the foundation system, especially that of fixed column bases, should be considered in the analysis of the structure and foundation systems.

Stability should be checked for the most critical overturning and sliding conditions. Care should be taken to ensure the stabilizing loads are not over estimated. For example, in high wind areas the empty load condition generally controls. For a rigid frame supported on two or more foundations, the stability of the entire system should be considered. Engineering judgment should be used to determine if the stability of the foundation system or an individual foundation within the system is more critical.

Air Cooler Components

Air coolers are supported by air cooler support beams, usually spanning longitudinally between the tops of the pipe rack columns. Project requirements will dictate actual elevation and framing details required to support the air coolers. When designing air cooler support beams, load application should envelope the design for maximum probable shear loads and bending moments. In cases where the air cooler support points do not match the width of the pipe rack, engineering judgment should be used to provide the optimum framing arrangement to support the air coolers.

Air cooler motors, and other components, require maintenance access. The access can be provided by walkways to the required maintenance locations (catwalks), or platforms sized to meet project requirements such as access to valve manifolds, access to air cooler platforms supplied with the equipment and support of smaller miscellaneous pieces of equipment. Catwalks can be provided by Fluor or the equipment supplier and can be supported by structural steel at the top level of the pipe rack hung from the air coolers.



PIPE SUPPORTS

Careful consideration should be given to support requirements for air cooler inlet and outlet piping, and flare header piping if it is located above the level of the air coolers.

PIPE BRIDGES

Pipe bridges are typically used when the spacing between support bents exceeds 40 feet (12m), which normally occurs at road crossings. Clearances over, under, and around pipe bridges are an important consideration in their design and are normally specified in the project design criteria.

Prior to starting a pipe bridge design, the Structural Engineer should verify with Piping Engineering where pipes will be supported on the bridge. As with pipe racks, pipe bridges can often be analyzed as multiple 2-D analyses models in lieu of 3-D analysis. Refer to Attachment 03.

Vertical Trusses

The vertical truss of a pipe bridge may be designed as a plane truss supporting gravity loads and in plane horizontal loads. In many cases, the vertical truss can be fabricated as a single shop welded unit of maximum dimensions based on shipping limitations. The connections of the truss should be designed to accommodate field assembly of the truss as individual members or as a unit. Cambering may be considered in accordance with the project design criteria and should be based on operating load deflection.

Detailing of the end connections between the vertical truss and the bridge bent should allow longitudinal movement of one of the chords so the end of the truss will behave as a pinned connection. Fully fixing both the top and bottom chords will prevent rotation at the end of the bridge and significantly affect the member sizes of both the chords and the bridge bent.

Horizontal Trusses

Horizontal trusses may be designed as plane trusses to resist all lateral loads applied to the truss such as wind or earthquake loads. Also, consideration should be given to providing lateral support to intermediate transverse beams, especially where anchors or large diameter pipes are present.

Bridge Bent

The bridge bent is designed similar to a typical pipe support frame with the exception that truss loads are applied as concentrated loads to the bent. Transverse vertical bracing should be used wherever access requirements allow.



PIPE SUPPORTS

The relative stiffness of large diameter pipes with respect to the bridge may be such that the gravity load of the pipe is partially redistributed to the supports at the bridge nets. In addition to considering these increased point loads in the design of the bridge bent, Piping Engineering and the Pipe Stress Engineer should be notified in order to properly consider these effects in the design of the pipe itself.

MISCELLANEOUS PIPE AND ELECTRICAL SUPPORTS

Miscellaneous supports required for pipes are MPS's and those required for cable tray or conduits are MES's. These include T-supports, sleeper supports, and other type of supports requested by Piping or Electrical Engineering. Refer to Attachment 03. These supports usually require a minimal amount of structural analysis; however, they often require a significant amount of effort to ensure that geometric constraints are satisfied.

Isolated small pipes, conduit and cable trays may be supported by cantilever beams from columns, or with hangers attached to beams.

T-Supports

T-supports are usually single columns with a short beam attached at top to support piping or electrical items. T-supports are usually cantilevered in both directions, however in some cases adequate support from piping or electrical trays or struts to adjacent structures can be used to provide support to the top of the column in one or both directions. Due to better bi-axial properties, tube sections can be a more economical alternative to wide flange sections and should be considered for T-supports.

Sleeper Supports

Sleeper supports are used to elevate pipes at low levels above the ground. Their design is relatively simple; however, close coordination with Piping and Pipe Stress Engineering is required to ensure that anchor loads are properly handled and settlement sensitive pipes are properly addressed.

Electrical Component Supports

Changes in cable tray direction and elevation may require significant steel framing to provide adequate support. Coordination with Electrical Engineering is required to ensure cable trays and conduits are properly supported.

Hold Down Supports

Hold down supports are a special type of support required to resist steady state dynamic loads applied to piping systems. These loads are typical of piping systems associated



PIPE SUPPORTS

with reciprocating compressors, but can result from any source of dynamic input. These supports are usually governed by the frequency and stiffness properties of the support.

Close coordination with Piping and Pipe Stress Engineering is required in order to ensure compatibility between the response of the supports and the piping system.

REVAMP OF EXISTING SUPPORTS

Considerations for revamp of pipe supports can be found in 000.200.0773 and 000.215.0965.

COST COMPETITIVE ENGINEERING AND DESIGN

Pipe supports should be designed in the most cost competitive manner possible. Cost competitive engineering and design of pipe supports meets the performance requirements of strength, stability, and serviceability and the project schedule requirements without unnecessary conservatism.

Additional guidance can be found in 000.200.1400 and 000.200.1401.

RESOURCES

Civil/Structural/Architectural Engineering:

[Specification 000.215.STI03310](#) Concrete Typical Details

[Drawing 000.215.5030](#) Structural Steel General Notes

[Drawing 000.215.5040](#) Structural Steel Connection Details (S1 thru S24)

[Drawing 000.215.5050](#) Structural Steel Connection Details (S30 thru S45)

[Drawing 000.215.5060](#) Structural Steel Connection Details (S52, Erection and Bracing Connections)

[Drawing 000.215.5230](#) Steel to Concrete - Typical Details

ACI 318/ACI318M American Concrete Institute - Building Code Requirements for Structural Concrete

ANSI/AISC 360 American Institute of Steel Construction - Specification for Structural Steel Buildings



PIPE SUPPORTS

Salmon and Johnson Steel Structures: Design and Behavior Emphasizing Load and Resistance Factor Design

REFERENCES

General Engineering:

[Practice 000.200.0773](#) Project Revamp Planning - Discipline Interface

[Practice 000.200.1400](#) Cost Competitive Engineering and Design

[Practice 000.200.1401](#) Design Margin Table

Civil/Structural/Architectural Engineering:

[Guideline 000.215.0760](#) Fireproofing Requirements for Structural Steel Members

[Guideline 000.215.0965](#) Guidelines for Field Investigations for Revamp Projects

[Guideline 000.215.1201](#) Single Plate Connections

[Guideline 000.215.1203](#) Bracing Connections

[Guideline 000.215.1208](#) Base Plate Design Criteria

[Guideline 000.215.1209](#) Bolted End Plate Moment Connections

[Guideline 000.215.1215](#) Wind Load Calculation

[Guideline 000.215.1216](#) Earthquake Engineering

[Guideline 000.215.STE05121](#) Anchor Bolt Design Criteria

[Specification 000.215.STC01015](#) Structural Design Guide

[Drawing 000.215.4060](#) Concrete Fireproofing of Structural Steel – Typical Details

[Drawing 000.215.4061](#) Pyrocrete Fireproofing of Structural Steel – Typical Details

[Reference 000.215.9885](#) Steel Connections - Details and Relative Costs

Piping Engineering:

[Reference 000.250.9821](#) Table of Weights - Piping Components



PIPE SUPPORTS

ASME B36.10M	Welded and Seamless Wrought Steel Pipe
ASTM C585	Standard Practice for Inner and Outer Diameters of Thermal Insulation for Nominal Sizes of Pipe and Tubing

ATTACHMENTS

Attachment 01	Weight of Piping and Insulation
Attachment 02	Torsion on Wide Flange Pipe Support Beams
Attachment 03	Typical Pipe Support Configurations
Attachment 04	Sample Design 1: Transverse Steel Rigid Frame Sample Design
Attachment 05	Sample Design 2: Piperack Strut and Brace Member Design
Attachment 06	Pipe Stress Interface Guidelines



PIPE SUPPORTS

Weight of Piping and Insulation

Table 1-1 Carbon Steel Pipe Weights - ASME B36.10M - NPS Designation (U.S. Customary Units in lbf/ft)

NPS	OD (in)	STD (Standard)				XS (Extra-Strong)				XXS (Double Extra-Strong)			
		t (in)	We (lbf/ft)	Ww (lbf/ft)	Wf (lbf/ft)	t (in)	We (lbf/ft)	Ww (lbf/ft)	Wf (lbf/ft)	t (in)	We (lbf/ft)	Ww (lbf/ft)	Wf (lbf/ft)
1	1.315	0.133	1.68	0.37	2.05	0.179	2.17	0.31	2.48	0.358	3.66	0.12	3.78
1-1/2	1.900	0.145	2.72	0.88	3.60	0.200	3.63	0.77	4.40	0.400	6.41	0.41	6.82
2	2.375	0.154	3.66	1.45	5.11	0.218	5.03	1.28	6.31	0.436	9.04	0.77	9.81
3	3.500	0.216	7.58	3.20	10.78	0.300	10.26	2.86	13.12	0.600	18.60	1.80	20.40
4	4.500	0.237	10.80	5.52	16.32	0.337	15.00	4.98	19.98	0.674	27.57	3.38	30.95
5	5.563	0.258	14.63	8.67	23.30	0.375	20.80	7.88	28.68	0.750	38.59	5.62	44.21
6	6.625	0.280	18.99	12.52	31.51	0.432	28.60	11.30	39.90	0.864	53.21	8.16	61.37
8	8.625	0.322	28.58	21.68	50.26	0.500	43.43	19.79	63.22	0.875	72.49	16.09	88.58
10	10.75	0.365	40.52	34.17	74.69	0.500	54.79	32.35	87.14	1.000	104.23	26.06	130.29
12	12.75	0.375	49.61	49.01	98.62	0.500	65.48	46.99	112.47	1.000	125.61	39.33	164.94
14	14.00	0.375	54.62	59.75	114.37	0.500	72.16	57.52	129.68	1.000*	138.97	49.01	187.98
16	16.00	0.375	62.64	79.15	141.79	0.500	82.85	76.58	159.43	1.000*	160.35	66.71	227.06
18	18.00	0.375	70.65	101.27	171.92	0.500	93.54	98.36	191.90	1.000*	181.73	87.13	268.86
20	20.00	0.375	78.67	126.12	204.79	0.500	104.23	122.86	227.09	1.000*	203.11	110.27	313.38
22	22.00	0.375	86.69	153.68	240.37	0.500	114.92	150.09	265.01	1.000*	224.49	136.14	360.63
24	24.00	0.375	94.71	183.97	278.68	0.500	125.61	180.04	305.65	1.000*	245.87	164.72	410.59
26	26.00	0.375	102.72	216.99	319.71	0.500	136.30	212.71	349.01	1.000*	267.25	196.04	463.29
28	28.00	0.375	110.74	252.72	363.46	0.500	146.99	248.11	395.10	1.000*	288.63	230.07	518.70
30	30.00	0.375	118.76	291.18	409.94	0.500	157.68	286.23	443.91	1.000*	310.01	266.83	576.84
32	32.00	0.375	126.78	332.36	459.14	0.500	168.37	327.07	495.44	1.000*	331.39	306.31	637.70
34	34.00	0.375	134.79	376.27	511.06	0.500	179.06	370.63	549.69	1.000*	352.77	348.51	701.28
36	36.00	0.375	142.81	422.89	565.70	0.500	189.75	416.92	606.67	1.000*	374.15	393.43	767.58
42	42.00	0.375	166.86	579.11	745.97	0.500	221.82	572.11	793.93	1.000*	438.29	544.54	982.83

For diameters, wall thicknesses and schedules not shown, refer to ASME B36.10M

For carbon steel piping component weights and relative weight factors for non-carbon steel piping materials, refer to Reference 000.250.9821

NPS	=	Nominal Pipe Size
OD	=	Outside Diameter
t	=	Wall Thickness
We	=	Empty Weight of Pipe
Ww	=	Weight of Water
Wf	=	Weight of Pipe Full of Water
*	=	No XXS Designation for the Given Line Size



PIPE SUPPORTS

Weight of Piping and Insulation

Table 1-2 Carbon Steel Pipe Weights - ASME B36.10M - DN Designation (SI Units in kgf/m)

DN	OD (mm)	STD (Standard)				XS (Extra-Strong)				XXS (Double Extra-Strong)			
		t (mm)	We (kgf/m)	Ww (kgf/m)	Wf (kgf/m)	t (mm)	We (kgf/m)	Ww (kgf/m)	Wf (kgf/m)	t (mm)	We (kgf/m)	Ww (kgf/m)	Wf (kgf/m)
25	33.4	3.38	2.50	0.56	3.06	4.55	3.23	0.46	3.69	9.09	5.45	0.18	5.63
40	48.3	3.68	4.05	1.31	5.36	5.08	5.40	1.14	6.54	10.15	9.54	0.61	10.15
50	60.3	3.91	5.45	2.16	7.61	5.54	7.49	1.90	9.39	11.07	13.45	1.14	14.60
80	88.9	5.49	11.28	4.77	16.05	7.62	15.27	4.26	19.53	15.24	27.68	2.68	30.36
100	114.3	6.02	16.07	8.21	24.28	8.56	22.32	7.41	29.74	17.12	41.03	5.03	46.06
125	141.3	6.55	21.77	12.90	34.67	9.53	30.95	11.73	42.69	19.05	57.43	8.36	65.79
150	168.3	7.11	28.26	18.63	46.89	10.97	42.56	16.81	59.37	21.95	79.19	12.15	91.33
200	219.1	8.18	42.53	32.26	74.79	12.70	64.63	29.45	94.08	22.23	107.88	23.94	131.82
250	273.0	9.27	60.30	50.85	111.15	12.70	81.54	48.15	129.7	25.40	155.11	38.78	193.89
300	323.8	9.53	73.83	72.93	146.76	12.70	97.44	69.93	167.4	25.40	186.93	58.53	245.46
350	355.6	9.53	81.28	88.92	170.20	12.70	107.39	85.60	193.0	25.40*	206.81	72.93	279.74
400	406.4	9.53	93.22	117.79	211.01	12.70	123.29	113.96	237.25	25.40*	238.63	99.27	337.90
450	457	9.53	105.14	150.71	255.85	12.70	139.20	146.37	285.58	25.40*	270.44	129.66	400.10
500	508	9.53	117.07	187.68	304.76	12.70	155.11	182.84	337.95	25.40*	302.26	164.10	466.36
550	559	9.53	129.01	228.71	357.72	12.70	171.02	223.36	394.38	25.40*	334.08	202.59	536.67
600	610	9.53	140.94	273.78	414.73	12.70	186.93	267.93	454.86	25.40*	365.89	245.14	611.03
650	660	9.53	152.86	322.91	475.78	12.70	202.84	316.55	519.39	25.40*	397.71	291.73	689.44
700	711	9.53	164.80	376.09	540.89	12.70	218.75	369.22	587.97	25.40*	429.53	342.38	771.91
750	762	9.53	176.73	433.33	610.06	12.70	234.65	425.95	660.60	25.40*	461.35	397.08	858.43
800	813	9.53	188.67	494.61	683.28	12.70	250.56	486.73	737.29	25.40*	493.16	455.83	949.00
850	864	9.53	200.59	559.95	760.54	12.70	266.47	551.56	818.03	25.40*	524.98	518.64	1043.6
900	914	9.53	212.52	629.33	841.86	12.70	282.38	620.44	902.82	25.40*	556.80	585.49	1142.3
1050	1067	9.53	248.32	861.81	1110.1	12.70	330.10	851.39	1181.5	25.40*	652.25	810.37	1462.6

For diameters, wall thicknesses and schedules not shown, refer to ASME B36.10M

For carbon steel piping component weights and relative weight factors for non-carbon steel piping materials, refer to Reference 000.250.9821

DN	=	Nominal Diameter
OD	=	Outside Diameter
t	=	Wall Thickness
We	=	Empty Weight of Pipe
Ww	=	Weight of Water
Wf	=	Weight of Pipe Full of Water
*	=	No XXS Designation for the Given Line Size



PIPE SUPPORTS

Weight of Piping and Insulation

Table 1-3 Carbon Steel Pipe Weights - ASME B36.10M - DN Designation (SI Units in N/m)

DN	OD (mm)	STD (Standard)				XS (Extra-Strong)				XXS (Double Extra-Strong)			
		t (mm)	We (N/m)	Ww (N/m)	Wf (N/m)	t (mm)	We (N/m)	Ww (N/m)	Wf (N/m)	t (mm)	We (N/m)	Ww (N/m)	Wf (N/m)
25	33.4	3.38	24.52	5.47	29.99	4.55	31.67	4.55	36.22	9.09	53.41	1.78	55.20
40	48.3	3.68	39.70	12.88	52.58	5.08	52.98	11.18	64.16	10.15	93.55	6.01	99.56
50	60.3	3.91	53.41	21.23	74.64	5.54	73.41	18.68	92.09	11.07	131.93	11.23	143.15
80	88.9	5.49	110.62	46.77	157.39	7.62	149.73	41.79	191.52	15.24	271.45	26.29	297.73
100	114.3	6.02	157.61	80.54	238.16	8.56	218.91	72.74	291.65	17.12	402.35	49.37	451.72
125	141.3	6.55	213.51	126.57	340.08	9.53	303.55	115.11	418.66	19.05	563.18	82.03	645.21
150	168.3	7.11	277.14	182.78	459.92	10.97	417.39	164.92	582.31	21.95	776.54	119.16	895.70
200	219.1	8.18	417.09	316.51	733.61	12.70	633.81	288.91	922.72	22.23	1057.9	234.87	1292.8
250	273.0	9.27	591.34	498.90	1090.2	12.70	799.60	472.38	1272.0	25.40	1521.1	380.45	1901.6
300	323.8	9.53	724.00	715.55	1439.6	12.70	955.61	686.05	1641.7	25.40	1833.1	574.24	2407.4
350	355.6	9.53	797.12	872.39	1669.5	12.70	1053.1	839.78	1892.9	25.40*	2028.1	715.55	2743.7
400	406.4	9.53	914.16	1155.6	2069.8	12.70	1209.1	1118.0	2327.2	25.40*	2340.1	973.94	3314.1
450	457	9.53	1031.1	1478.6	2509.7	12.70	1365.1	1436.1	2801.2	25.40*	2652.2	1272.1	3924.2
500	508	9.53	1148.1	1841.4	2989.5	12.70	1521.1	1793.8	3315.0	25.40*	2964.2	1610.0	4574.2
550	559	9.53	1265.1	2243.9	3509.0	12.70	1677.1	2191.4	3868.5	25.40*	3276.2	1987.6	5263.8
600	610	9.53	1382.2	2686.1	4068.3	12.70	1833.1	2628.7	4461.8	25.40*	3588.2	2405.0	5993.2
650	660	9.53	1499.1	3168.1	4667.2	12.70	1989.1	3105.7	5094.8	25.40*	3900.2	2862.2	6762.4
700	711	9.53	1616.1	3689.9	5306.0	12.70	2145.2	3622.5	5767.6	25.40*	4212.2	3359.1	7571.4
750	762	9.53	1733.2	4251.4	5984.6	12.70	2301.2	4179.0	6480.2	25.40*	4524.3	3895.8	8420.0
800	813	9.53	1850.2	4852.6	6702.9	12.70	2457.2	4775.3	7232.5	25.40*	4836.3	4472.2	9308.5
850	864	9.53	1967.1	5493.7	7460.8	12.70	2613.2	5411.4	8024.5	25.40*	5148.3	5088.4	10237
900	914	9.53	2084.2	6174.4	8258.6	12.70	2769.2	6087.2	8856.3	25.40*	5460.3	5744.3	11205
1050	1067	9.53	2435.1	8455.2	10890	12.70	3237.2	8353.1	11590	25.40*	6396.4	7950.6	14347

For diameters, wall thicknesses and schedules not shown, refer to ASME B36.10M

For carbon steel piping component weights and relative weight factors for non-carbon steel piping materials, refer to Reference 000.250.9821

DN	=	Nominal Diameter
OD	=	Outside Diameter
t	=	Wall Thickness
We	=	Empty Weight of Pipe
Ww	=	Weight of Water
Wf	=	Weight of Pipe Full of Water
*	=	No XXS Designation for the Given Line Size



PIPE SUPPORTS

Weight of Piping and Insulation

To determine the weight per foot (meter) of any piping insulation, use the Nominal Pipe Size (NPS) or Nominal Diameter (DN) and nominal insulation thickness to find the insulation cross-sectional area factor, F, in Tables 2-1 or 2-2 shown below. Then, multiply F by the density of the insulation in the appropriate units for the table being used.

Table 2-1 Insulation Weights - ASTM C585 - NPS Designation (U.S. Customary Units in lbf/ft)

Note: Average density of commonly used “perlite” insulation is 12 lbf/ft³, densities of other types of insulation may be found in Reference 000.250.9821.

Nominal Pipe Size (NPS)	Nominal Insulation Thickness								
	1"	1-1/2"	2"	2-1/2"	3"	3-1/2"	4"	4-1/2"	5"
1	0.057	0.101	0.159	0.229	0.307	0.396	0.495	0.621	0.743
1-1/2	0.067	0.116	0.219	0.297	0.385	0.485	0.610	0.733	0.867
2	0.079	0.137	0.207	0.285	0.374	0.473	0.599	0.721	0.855
3	0.101	0.171	0.249	0.337	0.437	0.562	0.685	0.819	1.001
4	0.127	0.205	0.293	0.393	0.518	0.641	0.775	0.957	1.115
5	0.143	0.232	0.331	0.457	0.580	0.713	0.896	1.054	1.223
6	0.160	0.260	0.385	0.508	0.642	0.824	0.982	1.151	1.331
8	---	0.340	0.474	0.656	0.814	0.983	1.163	1.354	1.556
10	---	0.429	0.587	0.757	0.937	1.127	1.329	1.542	1.766
12	---	0.497	0.677	0.868	1.070	1.282	1.506	1.741	1.986
14	---	0.493	0.684	0.886	1.099	1.322	1.557	1.802	2.059

Example: For NPS 4 pipe with 4" nominal insulation thickness, F = 0.775. If the insulation density is 12 lbf/ft³, then the insulation weight/ft is $0.775 \times 12 \text{ lbf/ft}^3 = 9.30 \text{ lbf/ft}$.



PIPE SUPPORTS

Weight of Piping and Insulation

Table 2-2 Insulation Weights - ASTM C585 - DN Designation (SI Units in kgf/m or N/m)

Note: Average density of commonly used “perlite” insulation is 192 kgf/m³ or 1883 N/m³, densities of other types of insulation may be found in Reference 000.250.9821.

Nominal Diameter (DN)	Nominal Insulation Thickness (mm)								
	25	38	51	64	76	89	102	114	127
25	0.005	0.009	0.015	0.021	0.029	0.037	0.046	0.058	0.069
40	0.006	0.011	0.020	0.028	0.036	0.045	0.057	0.068	0.081
50	0.007	0.013	0.019	0.027	0.035	0.044	0.056	0.067	0.080
80	0.009	0.016	0.023	0.031	0.040	0.052	0.063	0.076	0.093
100	0.012	0.019	0.027	0.036	0.048	0.059	0.072	0.089	0.104
125	0.013	0.022	0.031	0.042	0.054	0.066	0.083	0.098	0.113
150	0.015	0.024	0.036	0.047	0.060	0.077	0.091	0.107	0.124
200	---	0.031	0.044	0.061	0.076	0.091	0.108	0.126	0.145
250	---	0.040	0.055	0.070	0.087	0.105	0.124	0.143	0.164
300	---	0.046	0.063	0.081	0.100	0.119	0.140	0.162	0.184
350	---	0.046	0.065	0.083	0.102	0.122	0.145	0.167	0.192

Example: For DN 100 pipe with 102 mm nominal insulation thickness, $F = 0.072$. If the insulation density is 192 kgf/m³ or 1883 N/m³, then the insulation weight/m is $0.072 \times 192 \text{ kgf/m}^3 = 13.8 \text{ kgf/m}$ or $0.072 \times 1883 \text{ N/m}^3 = 136 \text{ N/m}$.

REFERENCES

Piping Engineering:

[Reference 000.250.9821](#) Table of Weights - Piping Components



TORSION ON WIDE FLANGE PIPE SUPPORT BEAMS

DETERMINATION OF TORSIONAL LOADS ON WIDE FLANGE BEAMS

Pipe support beams should be designed for torsional loads that are not effectively resisted by other mechanisms like counteracting vertical loads or the addition of plan bracing. Torsional loading conditions involving piping anchor and friction forces should consider the stabilizing influence of the piping weight. Considering anchor and friction forces alone results in unrealistic torsion stresses in the supporting beam. Considering the vertical piping weight in conjunction with the anchor and friction forces, the torsional stresses in the supporting beam are reduced.

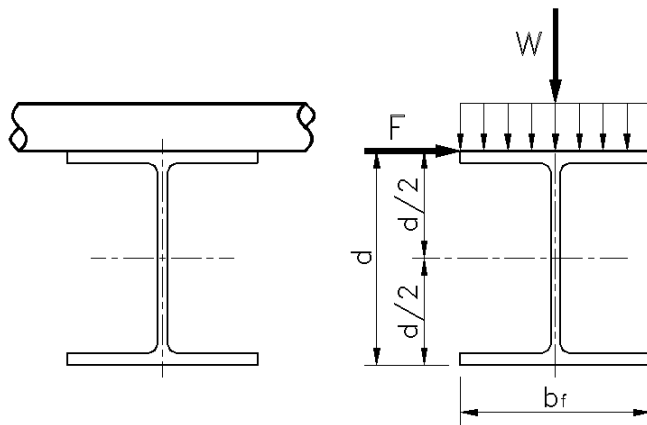


Figure 1
Applied Loads

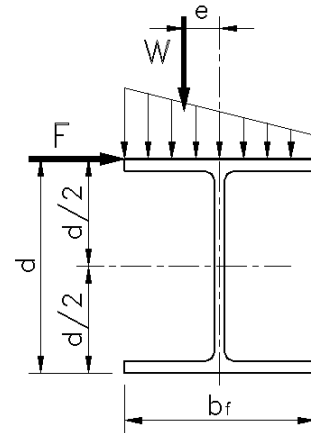


Figure 2
Small Lateral Load

Applied Loads

In Figure 1, F = the lateral anchor and friction force of the pipe applied at the top of the beam, and W = the weight of the pipe. For this analysis, it is assumed that the pipe is supported along the entire width of the top flange. For cases where the pipe is elevated above the top flange by a pipe shoe, the length of the pipe shoe should be used if it is less than the top flange width.

Small Lateral Loads

For small lateral loads, the effective location of the stabilizing weight (W) shifts a distance (e) to counteract the applied torsion. Figure 2 shows this loading with the applied torsion equal to the weight resisting torsion.

$$\frac{Fd}{2} = We$$

Where:

d = beam overall depth



TORSION ON WIDE FLANGE PIPE SUPPORT BEAMS

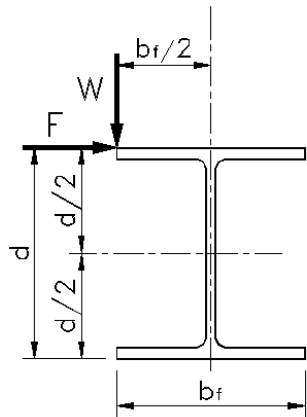


Figure 3
 Balanced Condition

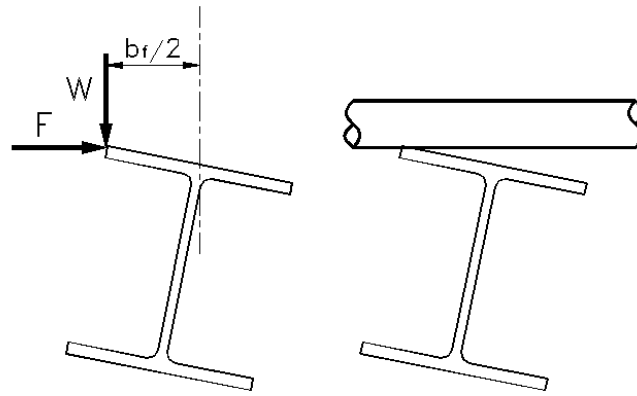


Figure 4
 Large Lateral Loads

Balanced Condition

As the lateral load magnitude increases, the effective location of the stabilizing weight (W) shifts until it reaches the end of the beam flange. Figure 3 shows this balanced loading condition when the applied torsion is equal to the maximum weight resisting torsion.

$$\frac{Fd}{2} = \frac{Wb_f}{2}$$

It follows that torsion on the beam does not occur when:

$$\frac{Fd}{2} \leq \frac{Wb_f}{2}$$

Large Lateral Loads

As the lateral load magnitude further increases, the stabilizing weight (W) can no longer counteract all of the applied torsion. The beam must be designed to resist a net applied torsion.

$$T_{net} = \frac{Fd}{2} - \frac{Wb_f}{2}$$

Figure 4 shows the case where the applied torsion exceeds the weight resisting torsion. A torsional analysis of the supporting beam is required considering only the net torsional load.



TORSION ON WIDE FLANGE PIPE SUPPORT BEAMS

ANALYSIS OF TORSIONAL LOADS ON WIDE FLANGE BEAMS

General

Torsional loads cause warping torsional stresses along the length of the beam flanges. These torsional stresses must be added to the strong-axis flexural stresses and weak-axis flexural stresses. All of these stresses act as axial stresses along the length of the beam flanges. The summation of these stresses needs to be compared with the acceptable capacity prescribed in the applicable code.

$$f = f_{bx} + f_{by} + f_{bw}$$

Where:

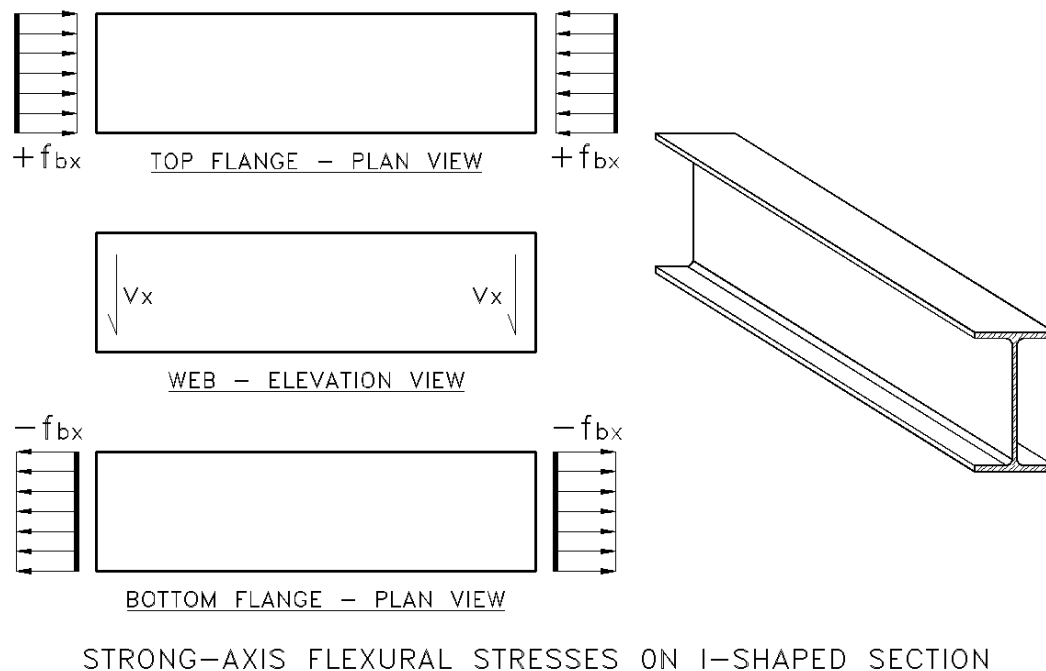
f = total flexural stress

f_{bx} = flexural stress resulting from strong-axis (x-axis) loading

f_{by} = flexural stress resulting from weak-axis (y-axis) loading

f_{bw} = flexural stress resulting from torsional loading

Strong-Axis Flexural Stresses





TORSION ON WIDE FLANGE PIPE SUPPORT BEAMS

Strong-axis flexural stresses (f_{bx}) are determined in accordance with standard methods.

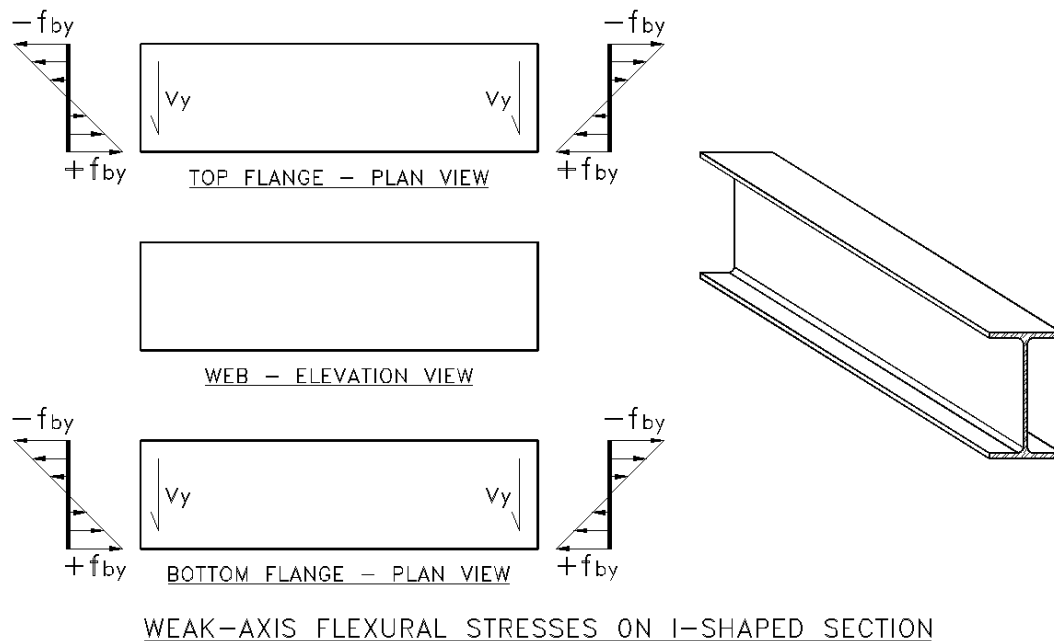
$$f_{bx} = \frac{M_x}{S_x}$$

Where:

M_x = bending moment resulting from strong-axis (x-axis) loading

S_x = elastic section modulus referenced to strong-axis (x-axis)

Weak-Axis Flexural Stresses



Weak-axis flexural stresses (f_{by}) are determined in accordance with standard methods.

$$f_{by} = \frac{M_y}{S_y}$$

Where:

M_y = bending moment resulting from weak-axis (y-axis) loading

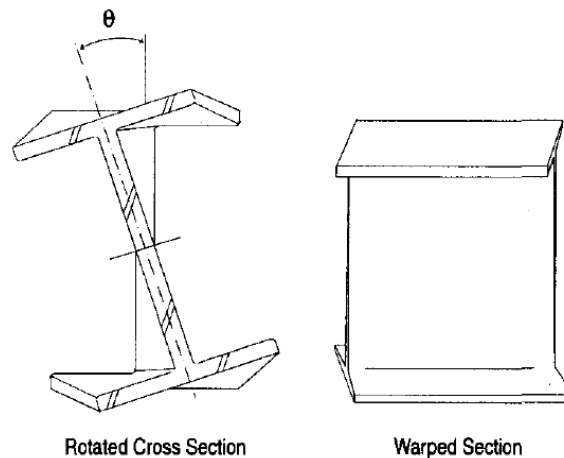
S_y = elastic section modulus referenced to weak-axis (y-axis)



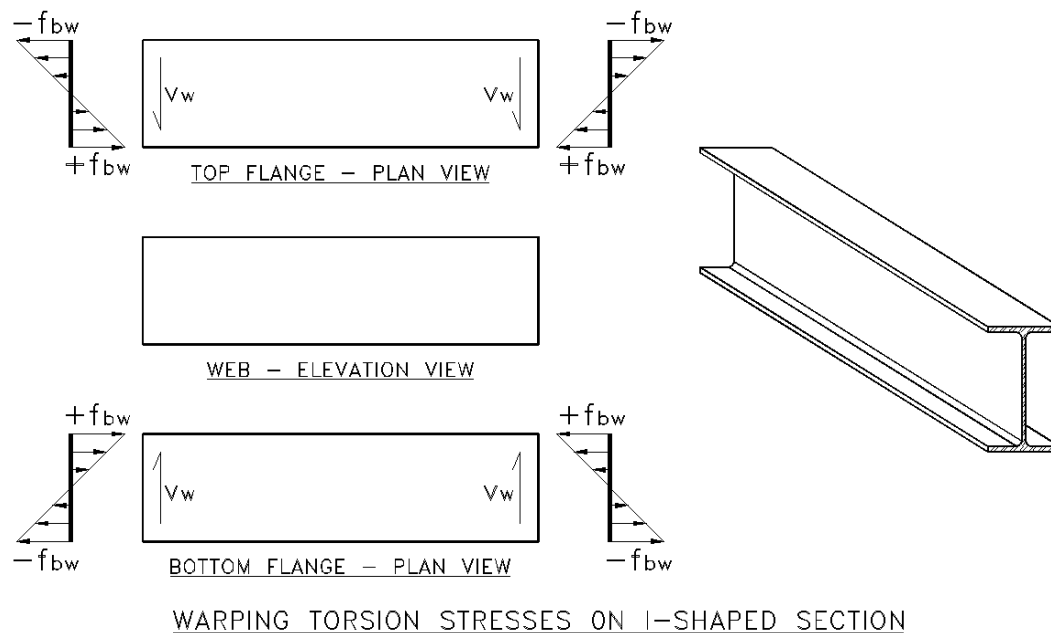
TORSION ON WIDE FLANGE PIPE SUPPORT BEAMS

Warping Torsion Stresses

Warping torsion is the out-of-plane effect that arises when the beam flanges twist and are laterally displaced in response to a torsional load.



Warping torsional stresses are flexural stresses (tensile and compressive) that act perpendicular to the surface of the cross-section and are constant across the thickness of an element of the cross-section but vary in magnitude along the length of the element.





TORSION ON WIDE FLANGE PIPE SUPPORT BEAMS

Warping torsional stresses (f_{bw}) are evaluated between points of lateral support for pipe support beams, usually the ends of the beam and work points of horizontal bracing. Precise solutions for the warping torsion stresses involves differential equations and can be found in many published sources. See References.

$$f_{bw} = \frac{Eb_f h}{4} \frac{d^2 \phi}{dz^2}$$

Where:

E = beam material modulus of elasticity (Young's modulus)

b_f = beam flange width

h = distance between center of beam flanges

$\frac{d^2 \phi}{dz^2}$ = flange moment per unit length

θ = beam twist angle

z = beam longitudinal axis

Approximate solutions for the warping torsion stresses are also available, including treating the lateral load as if it is resisted by half of the cross section. These approximate methods may give either unconservative or excessively conservative results. Their use should be discouraged.

References

American Institute Of Steel construction (AISC), Design Guide 9, Torsional Analysis Of Structural Steel Members, 2003.

Salmon, Charles G, John E. Johnson, and Faris A Malhas, Steel Structures Design And Behavior, 5th Edition, 2009.

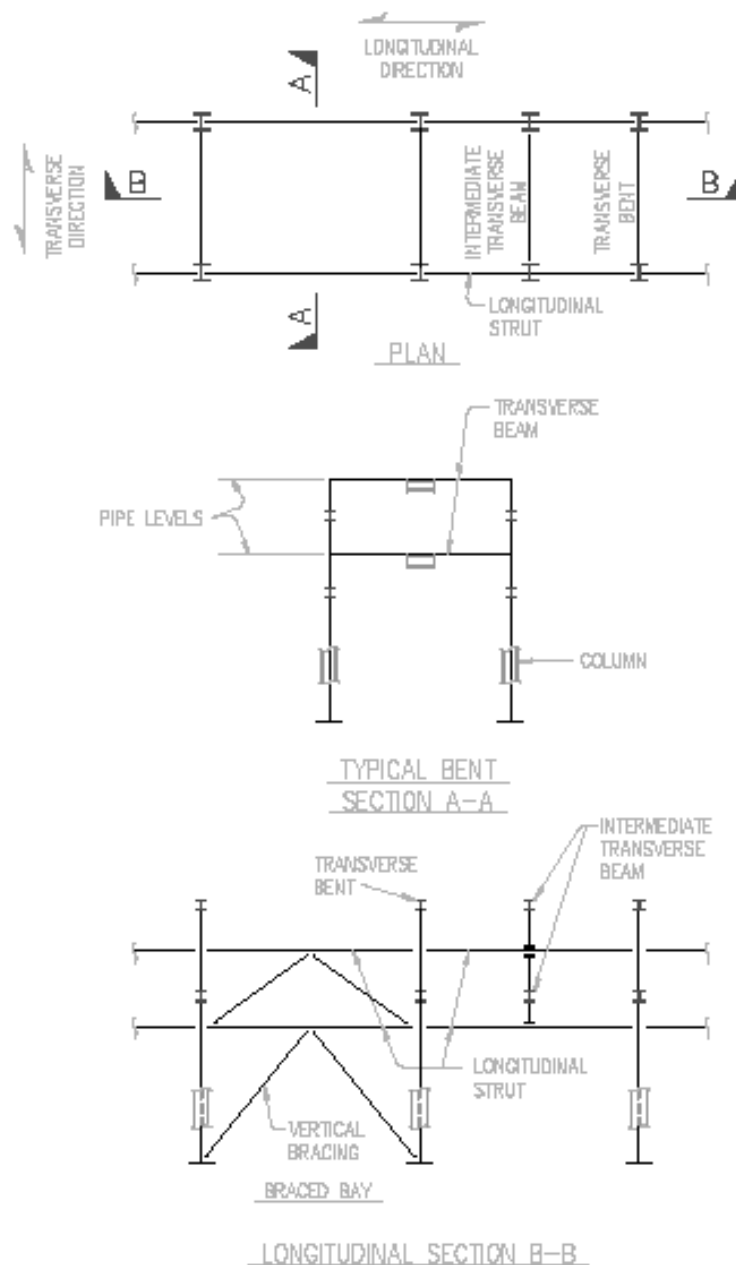
Young, W.C. and R.G. Budynas, Roark's Formulas For Stress And Strain, 7th Edition, 2002.



PIPE SUPPORTS

Typical Pipe Support Configurations

Typical Pipe Rack Configuration



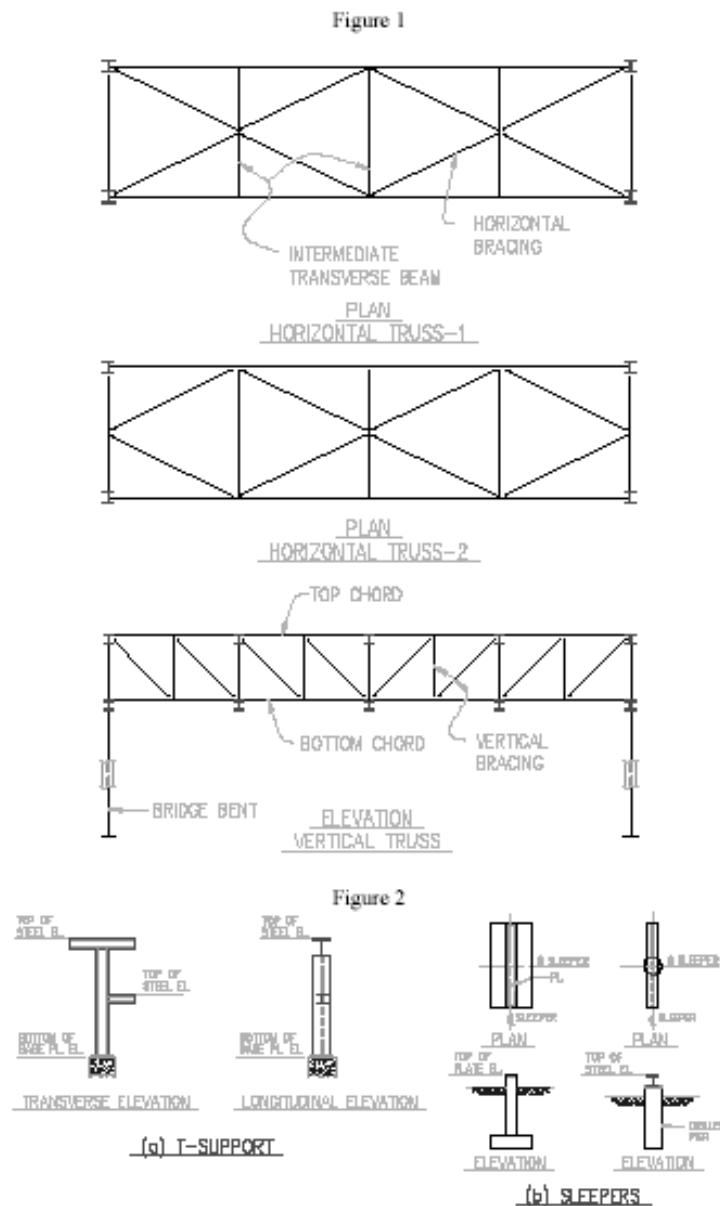


PIPE SUPPORTS

Typical Pipe Support Configurations

Figure 1 – Typical Pipe Bridge

Figure 2 – Miscellaneous Pipe Supports

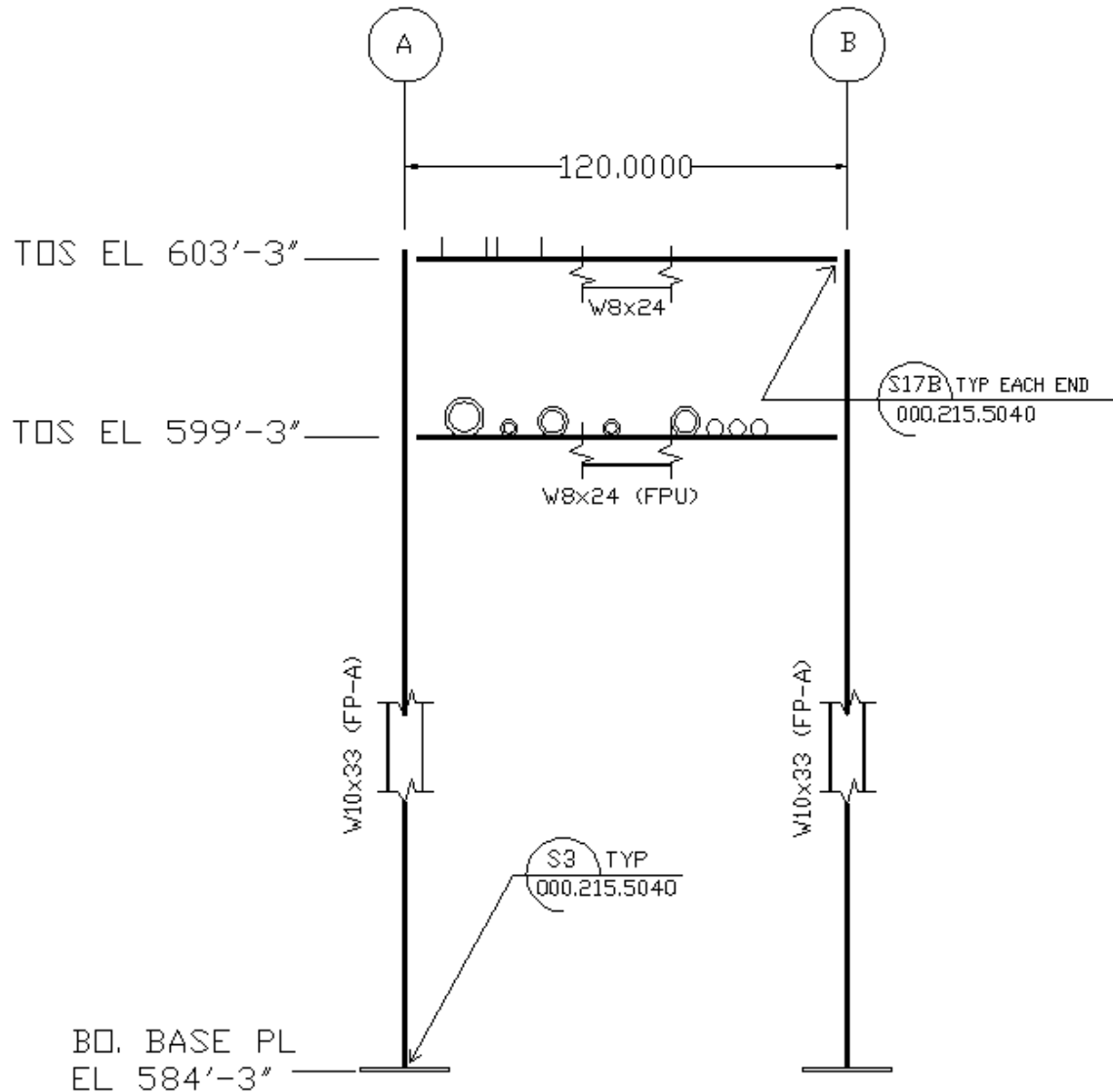




SAMPLE DESIGN 1: TRANSVERSE STEEL RIGID FRAME SAMPLE DESIGN

SKETCH

The transverse pipe bent shown represents a small pipeway on a US Project.





SAMPLE DESIGN 1: TRANSVERSE STEEL RIGID FRAME SAMPLE DESIGN

GIVEN

References

- AISC Manual of Steel Construction, 13th Edition

Material

- Structural Steel – ASTM A992

Loads

Loads are determined in accordance with the project-specific Structural Design Criteria document.

Operating dead load of pipe racks is prescribed as a uniformly distributed load of 40psf.

$$(40 \text{ psf})(16 \text{ ft}) = 640 \text{ lb / ft}$$

Empty dead load is taken to be 60% of the assumed operating dead load of the pipe.

$$0.60(640 \text{ lb / ft}) = 384 \text{ lb / ft}$$

Electrical cable tray dead load @TOS EL. 603'-3"

$$(20 \text{ psf})(16 \text{ ft}) = 320 \text{ lb / ft}$$

Fireproofing dead load

161 plf on columns

100 plf on beam @ TOS EL. 599'-3"

Wind load calculations, performed in accordance with Guideline 000.215.1215, are not shown for brevity.

Wind on cable tray @TOS EL. 603'-3" is 740 lb

Wind on piping @TOS EL. 599'-3" is 780 lb

Wind load on column is 16 plf

Earthquake load calculations, performed in accordance with Guideline 000.215.1216, are not shown for brevity. The loads were determined to be much less than the wind loads and are not summarized here.

$$T = 0.353 \text{ sec}$$

In lieu of anchor load information from piping, a value of 2 kips was applied at the mid span of the beam.

$$\text{Thermal component: } 0.30(2 \text{ K}) = 0.6 \text{ K}$$

$$\text{Friction component: } 0.70(2 \text{ K}) = 1.4 \text{ K}$$

Load Combinations

Load combinations are determined in accordance with the project-specific Structural Design Criteria document. Strength (LRFD) load combinations are taken from the project-specific Structural Design Criteria document, which is based on ASCE 7-05 and PIP STS05120.



SAMPLE DESIGN 1: TRANSVERSE STEEL RIGID FRAME SAMPLE DESIGN

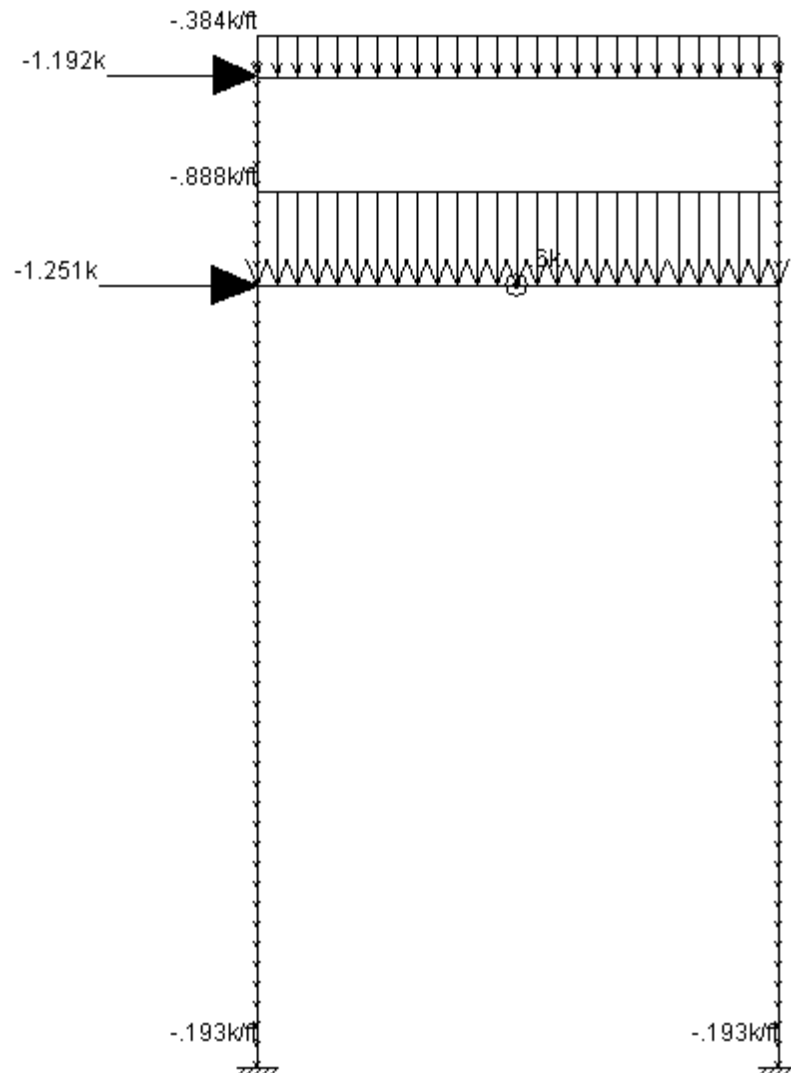
REQUIRED

Determine steel member sizes for the given loading conditions.

SOLUTION

Load Combinations

The dominant load combination for the beam and column members was determined from a RISA analysis to be $1.2D + 1.6W + A_f$





SAMPLE DESIGN 1: TRANSVERSE STEEL RIGID FRAME SAMPLE DESIGN

Beam @ TOS EL. 599'-3"

Required strengths: $1.2D + 1.6W + A_f$

$$P_u = 1.438 \text{ K} \quad \{\text{from RISA analysis}\}$$

$$V_{ux} = 6.07 \text{ K at end} \quad V_{ux} = 1.487 \text{ K at middle} \quad \{\text{from RISA analysis}\}$$

$$M_{ux} = 14.382 \text{ ft-K at end} \quad M_{ux} = 5.711 \text{ ft-K at middle} \quad \{\text{from RISA analysis}\}$$

$$M_{uy} = 0.000 \text{ ft-K at end} \quad M_{uy} = 1.453 \text{ ft-K at middle} \quad \{\text{from RISA analysis}\}$$

Check W8x24

$$L_b = 10 \text{ ft} / 3 = 3.33 \text{ ft} \quad \{\text{unbraced length of compression flange}\}$$

$$L_p = 5.69 \text{ ft} \quad \{\text{from Steel Manual}\}$$

$$L_b \leq L_p \quad \text{no lateral torsional buckling}$$

$$\phi_b M_{nx} = \phi_b M_{px} = 86.6 \text{ ft-K} \quad \{\text{from Steel Manual}\}$$

$$\phi_b M_{ny} = \phi_b M_{py} = 32.1 \text{ ft-K} \quad \{\text{from Steel Manual}\}$$

$$\phi_v V_{nx} = 58.3 \text{ K} \quad \{\text{from Steel Manual}\}$$

$$KL \approx 0.65(10 \text{ ft}) = 6.5 \text{ ft} \quad \{\text{approximate effective length}\}$$

$$\phi_c P_n = 268.1 \text{ K} \quad \{\text{from Steel Manual}\}$$

$$\frac{V_{ux}}{\phi_v V_{nx}} = \frac{6.07 \text{ K}}{58.3 \text{ K}} = 0.10 \leq 1.00 \quad \text{ok}$$

$$\frac{P_u}{2\phi_c P_n} + \left(\frac{M_{ux}}{\phi_b M_{nx}} + \frac{M_{uy}}{\phi_b M_{ny}} \right) = \frac{1.438 \text{ K}}{2(268.1 \text{ K})} + \left(\frac{14.382 \text{ ft-K}}{86.6 \text{ ft-K}} + 0 \right) = 0.17 \leq 1.0 \quad \text{ok}$$

Use W8x24 at TOS EL. 599'-3".

Loads on TOS EL. 603'-3" are less than TOS EL. 599'-3". Use W8x24 on both levels



SAMPLE DESIGN 1: TRANSVERSE STEEL RIGID FRAME SAMPLE DESIGN

Column

Required strengths: $1.2D + 1.6W + A_f$

$$P_u = 8.771 K \text{ left column} \quad P_u = 13.445 K \text{ right column} \quad \{\text{from RISA analysis}\}$$

$$V_{ux} = 1.542 K \text{ left column} \quad V_{ux} = 2.848 K \text{ right column} \quad \{\text{from RISA analysis}\}$$

$$M_{ux} = 13.035 \text{ ft} - K \text{ left column} \quad M_{ux} = 14.688 \text{ ft} - K \text{ right column} \quad \{\text{from RISA analysis}\}$$

$$M_{uy} = 0.000 \text{ ft} - K \quad \{\text{loads taken by longitudinal frame}\}$$

Check W10x33

$$L_b = 15 \text{ ft} \quad \{\text{unbraced length of compression flange}\}$$

$$L_p = 6.85 \text{ ft} \quad \{\text{from Steel Manual}\}$$

$$L_b > L_p \quad \underline{\text{consider lateral torsional buckling}}$$

$$C_b \approx 1.67 \quad \{\text{for approximately triangular moment diagram}\}$$

$$\phi_b M_{px} = 146 \text{ ft} - K \quad \{\text{from Steel Manual}\}$$

$$\begin{aligned} \phi_b M_{nx} &= (1.67)(116 \text{ ft} - K) \leq 146 \text{ ft} - K \\ &= 194 \text{ ft} - K \leq 146 \text{ ft} - K \Rightarrow = 146 \text{ ft} - K \end{aligned} \quad \{\text{from Steel Manual}\}$$

$$\phi_v V_{nx} = 84.7 K \quad \{\text{from Steel Manual}\}$$

$$(KL)_y = 1.0(15 \text{ ft}) = 15.0 \text{ ft} \quad \{\text{controlling effective length}\}$$

$$\phi_c P_n = 253 K \quad \{\text{from Steel Manual}\}$$

$$\frac{V_{ux}}{\phi_v V_{nx}} = \frac{2.848 K}{84.7 K} = 0.03 \leq 1.00 \quad \underline{\text{ok}}$$

$$\frac{P_u}{2\phi_c P_n} + \left(\frac{M_{ux}}{\phi_b M_{nx}} + \frac{M_{uy}}{\phi_b M_{ny}} \right) = \frac{13.445 K}{2(253 K)} + \left(\frac{14.688 \text{ ft} - K}{146 \text{ ft} - K} + 0 \right) = 0.13 \leq 1.0 \quad \underline{\text{ok}}$$

Use W10x33

Other

Connections, base plates, and anchor bolts calculations, performed in accordance with Guidelines 000.215.1208, 000.215.1209 and 000.215.STE05121, are not shown for brevity.

The intent of this example is to demonstrate a general methodology of strut and bracing design using Canadian code based loads and load combinations. The Canadian codes are a nationally coordinated set of documents for design. The load combination philosophy is presented in regionally enacted Building Codes. The National Building Code of Canada (NBCC) is the template that is used for all regional codes. The regional Building Code may simply be the legislated implementation of the NBCC as is or be a separate document developed to include minor regional additions and modifications.

Struts are primarily designed to transfer any longitudinal loads along the length of the piperack to the bracing, as well as support any vertical loading that may be required. Bracing is designed to transfer the cumulative longitudinal loads from the struts, down to the foundation. The struts and bracing also provide points of stability for the columns.

Consider a piperack with 2 levels of longitudinal piping and the geometry shown below:

BOUNDARY CONDITIONS:

All struts and bracing are considered to be "pin" end connected.

BASIC LOAD CASES:

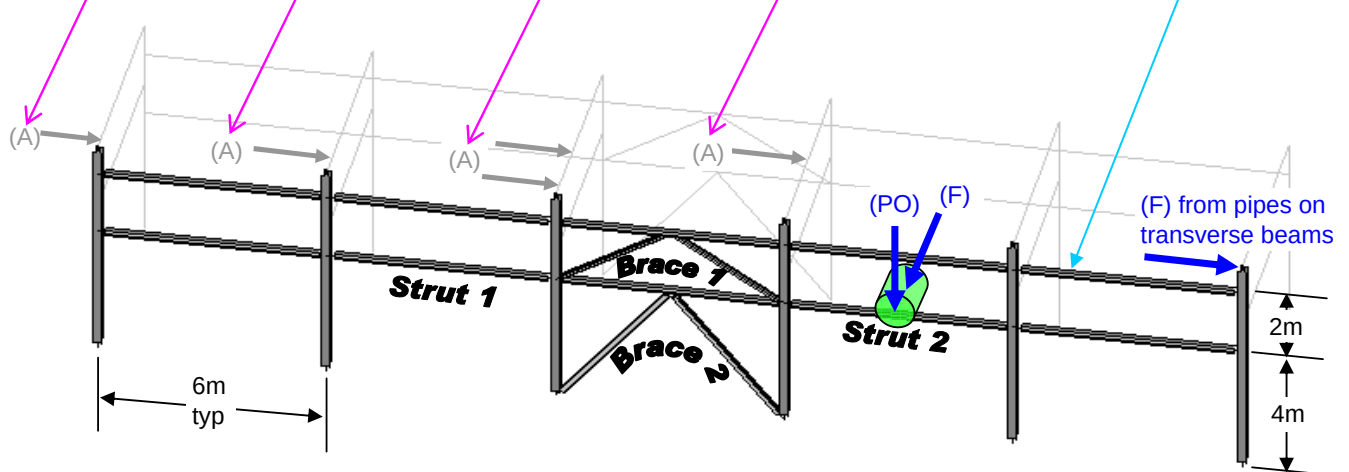
Define some of the potential loads that act on this strut and brace system:

- (SW) Self Weight - steel, connections, fireproofing and miscellaneous dead loads.
- (PO) Pipe Operating
- (PE) Pipe Empty - will likely not govern as a principal load on a strut or brace.
- (S) Snow - will likely not govern as a principal load on a stringer or brace.
- (WT) Wind Transverse - will likely not govern as a principal load on a stringer or brace.
- (WL) Wind Longitudinal - generally neglected; longitudinal friction and anchor loads typically govern the design.
- (EQT) Earthquake Transverse - will likely not govern as a principal load on a stringer or brace.
- (EQL) Earthquake Longitudinal
- (F) Pipe Friction - load transfers from transverse beams -> columns -> struts -> bracing -> foundation.
- (AT, AF, AE) Pipe Anchor - Anchor Thermal, Anchor Friction, Anchor Earthquake.
- (T) Temperature - ambient temperature loads. Becomes increasingly significant with broader temperature ranges, further distances between end frame and braced bay, and closer distances between the lowest strut level and grade. This only occurs when there is more than one level of struts.

**PIPERACK STRUT AND BRACE MEMBER DESIGN EXAMPLE
USING CANADIAN BUILDING CODE LOAD COMBINATIONS**

Note: It is not appropriate to apply assumed anchor loads used for local transverse beam design as the strut and brace design load. All final beam anchor loads are unlikely to be acting simultaneously at full value in the same direction.

Note: struts can potentially be deleted if they are not supporting pipe or are not part of the column stability system.



LOAD COMBINATIONS:

The basic load cases defined above must be considered as either a principal or companion load. A principal load is the dominant load of a particular combination and is assigned a load factor to account for the variability of its effects or confidence of its magnitude. A companion load is a secondary load which accompanies the principal load and is assigned a load factor representing its probable magnitude of acting simultaneously with the principal load. The table below represents some of the load combinations which may potentially govern the member design.

COMBINATION	PERMANENT	PRINCIPAL	COMPANION
1	1.4(SW + PO)		
2	1.25(SW + PO + AT)		0.5S
3	1.0(SW + PO + AT)	1.0(AE + EQL)	0.25S
4	1.25(SW + PO + AT)	1.25F	0.25(AF + WT)
5	1.25(SW + PO + AT)	1.25AF	0.25(F + WT)
6	1.25(SW + PO + AT)	1.25F	0.4WT

STRUT DESIGN:

STRUT 1:

This is the largest axially loaded 6m long strut. Typically a W-Shape section is most appropriate as a strut. The primary load of concern will be the axial compressive loads on the strut. It is likely that slenderness limitations will govern this strut selection before strength capacity.

--> select a member that is light in weight with a satisfactory " λ " value such that $KL/r < 200$. A rule of thumb for a strut would be to have the strut depth be approximately equal to span/30.

--> use the "factored axial compressive resistance" tables in the CISC steel handbook and compare the selected member axial resistance to the factored compressive load of Strut 1. This will quickly identify likely member choices; however, detailed code checks are required by the engineer for final design.

STRUT 2:

In addition to the axial loading from Strut 1, Strut 2 has bending about both axis. Begin with the member size chosen for Strut 1 as a starting point. It is likely that Strut 2 will need to be deeper and heavier as member strength, rather than slenderness, may be the governing design criteria.

--> follow steps for strut 1 above

--> use the "beam selection tables" in the CISC steel handbook and compare the selected member bending resistance to the factored moments for Strut 2 in both the strong and weak axis.

--> apply simplified biaxial bending formula $C_r/C_r + M_{fx}/M_{rx} + M_{fy}/M_{ry} \leq 1.0$ iterate as needed. This will quickly identify likely member choices; however, detailed final code checks are required by the engineer.

--> check if pipe stress has vertical or lateral deflection concerns and check deflections as required. If lateral deflection is a concern for anchors on struts it is important to include the transverse frame deflection when determining deflection of the anchor location.

STRUT CONNECTION DESIGN:

End reactions for struts would typically be communicated on the design drawings for connection design to be done by the fabricator.

BRACE DESIGN:

SELECT BRACE STYLE:

- X-Brace (Cross Brace) - used when access across braced area is not required, and for large loads and reasonably square geometry, requires connection at the crossing of the bracing.
- V-Brace (Chevron) - used for long spans, large loads, rectangular geometry, requires 3 connection points.
- Z-Brace (Single brace) - used for shorter spans, requires 2 connection points.
 - > use Chevron style bracing since it provides some access while accommodating the long span and geometry of the braced bay. Brace angles sharper than 3:1 can lead to unreasonable gusset sizes.

SELECT BRACE MEMBER SHAPE:

- W-Shape - used for large loads and long spans, available in many sizes, connections are concentric.
- WT - used for medium loads and medium spans, available in many sizes connections are eccentric.
- 2L-Back to Back - efficient in load/weight ratios, double shear bolt connections, connection are eccentric.
- 2L-Star Shaped - efficient in load/weight ratios, connections are concentric, capacity tables in CISC handbook.
- L-Shape - used for light loads and short spans. Connections are eccentric.
 - > use a 2L shape for their efficiency.

BRACE 1:

This brace set collects the cumulative axial load from the top stringers.

--> select a 2L set that is light in weight with a satisfactory "r" value such that $KL/r < 200$. The r_x and r_y values should be comparable in magnitude for an efficient design.

--> use the "factored axial compressive resistance" tables for double angle struts in the CISC steel handbook and compare the selected member axial resistance to the factored compressive load of Brace 1. This will identify likely member choices while keeping in mind the capacity reduction due to the style and spacing of interconnection points. Detailed code checks are required by the engineer.

BRACE 2:

This brace set collects the cumulative axial load from the top and bottom stringers..

--> follow steps for Brace 1, noting that there may be a need to increase the member size and/or weight or even change the member shape to account for the increase in load and the longer length of the brace.

BRACE CONNECTION DESIGN:

End connections for bracing would typically be done in-house but are not within the scope of this example. An option is to provide the appropriate basic load cases and load combinations for the fabricator to be able to cost-effectively design the connections.

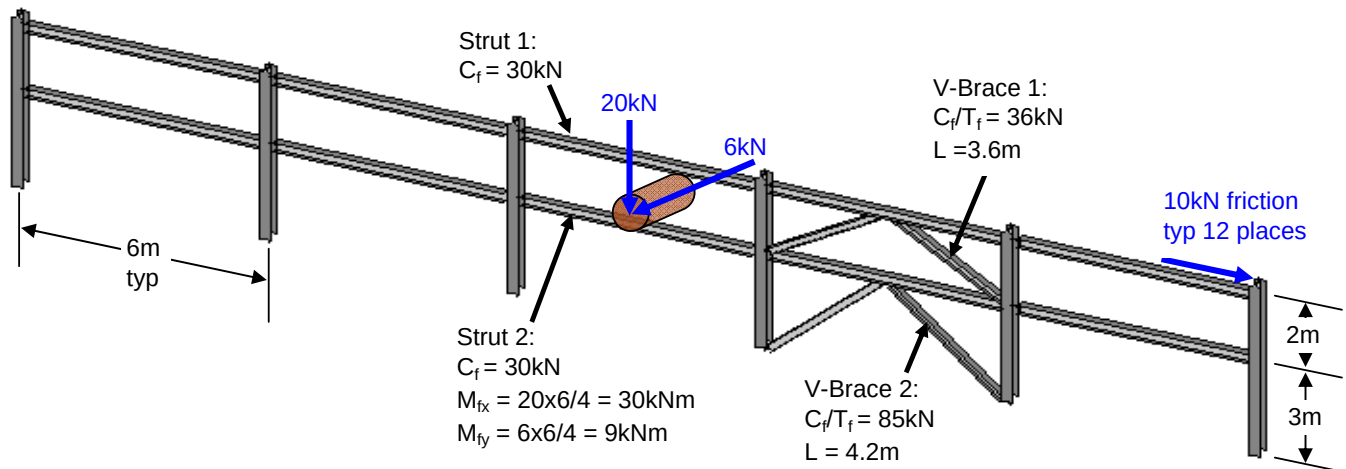
PIPERACK STRUT AND BRACE MEMBER DESIGN EXAMPLE USING CANADIAN BUILDING CODE LOAD COMBINATIONS

Consider a 5 bay piperack with 2 levels of longitudinal piping and the geometry shown below.

This braced bay shall be designed to transfer the pipe frictional forces from the transverse beams (not shown) through the longitudinal struts and vertical bracing to the foundation. In addition, one of the longitudinal struts provides vertical support for a single pipe as shown.

All members in the longitudinal load resisting system are pin end connected.

Load development, load combinations, connection design etc are outside the scope of this example.



LOADING:

All loads shown or calculated are considered as factored loads.

Friction loads are assumed to be simultaneously acting in the same direction, and are reversible. Assume a 10kN friction load is transferred from each of the transverse beams where shown above.

Pipe loads are shown above for a single pipe which turns out of the piperack and is supported at midspan of a strut. Assume a 20kN vertical load and a 6kN frictional load as shown.

Other load types and scenarios can act on a strut and V-brace system that are not considered in this example.

STRUT DESIGN:

Strut 1:

This is the largest axially loaded 6m long strut. The cumulative axial load is $3 \times 10\text{kN} = 30\text{kN}$

Check the slenderness ratio such that $KL/r < 200$:

$$(1.0 \times 6000) / r < 200 \rightarrow r_{\min} = 30$$

$$r_{y,W200 \times 27} = 31.2 > 30 \quad \text{OK}$$

**PIPERACK STRUT AND BRACE MEMBER DESIGN EXAMPLE
USING CANADIAN BUILDING CODE LOAD COMBINATIONS**

Check compressive resistance of W200x27 at 6m length:

$$C_r, W200x27 @ 6m = 154kN \text{ (from CISC handbook column tables)}$$

$$C_r/C_r = 30/154 = 0.19 < 1.0 \text{ OK}$$

Use W200x27 for member 1

Strut 2:

In addition to the axial loading from member 1, this member must be designed for biaxial bending due to the pipe loads at midspan.

Try the W200x27 as used for Strut 1:

Find the member class:

$$b/t = (133/2)/(8.4) = 7.92 = \text{class 2}$$

$$\rightarrow C_r/C_r + 0.85U_{1x}M_{fx}/M_{rx} + \beta U_{1y}M_{fy}/M_{ry} < 1.0 \text{ (CSA S16-01, 13.8.2)}$$

Check cross sectional strength as per S16-09 13.8.2a:

$$C_r = C_{r,max} = 1050kN \text{ (from CISC handbook column tables)}$$

$$M_{rx} = M_r = 86.6kN \text{ (from CISC handbook column tables)}$$

$$M_{ry} = 23.6kN \text{ (from CISC handbook column tables)}$$

$$C_{ex} = \pi^2 EI_x / L^2 = \pi^2 (200000)(25.8 \times 10^6) / (6000)^2 = 1.4 \times 10^6 kN$$

$$C_{ey} = \pi^2 EI_y / L^2 = \pi^2 (200000)(3.30 \times 10^6) / (6000)^2 = 0.18 \times 10^6 kN$$

$$\omega_1 = 1.0 \text{ (for pin ended members)}$$

$$U_{1x} = \omega_1 / (1 - C_r/C_{ex}) = 1.02$$

$$U_{1y} = \omega_1 / (1 - C_r/C_{ey}) = 1.20$$

$$\beta = 0.6 \text{ (0.6 for 13.8.2a check only)}$$

$$\rightarrow C_r/C_r + 0.85U_{1x}M_{fx}/M_{rx} + \beta U_{1y}M_{fy}/M_{ry} < 1.0$$

$$\rightarrow 30/1050 + 0.85(1.02)(30/86.6) + 0.6(1.20)(9/23.6) < 1.0$$

$$\rightarrow 0.03 + 0.30 + 0.28 = 0.61 < 1.0 \text{ OK}$$

Check overall member strength as per S16-09 13.8.2b:

$$F_{ex} = \pi^2 E / (K_x L_x / r_x)^2 = \pi^2 (200000) / (1.0 \times 6000 / 87.3)^2 = 418MPa$$

$$\lambda_x = \sqrt{(F_y / F_{ex})} = \sqrt{(350 / 418)} = 0.915$$

$$C_r = C_{rx} = \Phi A F_y (1 + \lambda_x^{2n})^{-1/n} = 0.9 \times 3390 \times (1 + 0.915^{2 \times 1.34})^{-1/1.34} = 692kN$$

PIPERACK STRUT AND BRACE MEMBER DESIGN EXAMPLE

USING CANADIAN BUILDING CODE LOAD COMBINATIONS

$$\lambda_y = (KL/r_y) \sqrt{(F_y/\pi^2 E)} = (1.0 \times 6000/31.2) \sqrt{(350/(\pi^2 \times 200000))} = 2.56$$

$$\beta = 0.6 + 0.4\lambda_y = 0.6 + 0.4(2.56) = 1.62$$

$$\rightarrow C_r/C_r + 0.85U_{1x}M_{fx}/M_{rx} + \beta U_{1y}M_{fy}/M_{ry} < 1.0$$

$$\rightarrow 30/692 + 0.85(1.02)(30/86.6) + 1.62(1.20)(9/23.6)$$

$$\rightarrow 0.04 + 0.30 + 0.74 = 1.08 > 1.0 \text{ NG}$$

Check lateral torsional buckling strength as per S16-09 13.8.2c:

$$C_r @ 6m = 154kN \text{ (from CISC handbook column tables)}$$

$$M_{rx} @ 6m = 32.6kN \text{ (from CISC handbook beam selection tables)}$$

$$\rightarrow C_r/C_r + 0.85U_{1x}M_{fx}/M_{rx} + \beta U_{1y}M_{fy}/M_{ry} < 1.0$$

$$\rightarrow 30/154 + 0.85(1.02)(30/32.6) + 1.62(1.20)(9/23.6)$$

$$\rightarrow 0.19 + 0.80 + 0.74 = 1.73 > 1.0 \text{ NG}$$

*must consider a heavier and potentially deeper member to control major and minor axis bending and to limit the deflection of the member under piping loads.

Try a W310x39 for Strut 2:

Follow the procedure above. Summary of the checks of 13.8.2 for a W310x39 are shown below:

Check cross sectional strength as per S16-09 13.8.2a:

$$\rightarrow 30/1556 + 0.85(1.01)(30/189) + 0.6(1.08)(9/42.5)$$

$$\rightarrow 0.02 + 0.14 + 0.14 = 0.30 < 1.0 \text{ OK}$$

Check overall member strength as per S16-09 13.8.2b:

$$\rightarrow 30/1305 + 0.85(1.01)(30/189) + 1.43(1.08)(9/42.5)$$

$$\rightarrow 0.02 + 0.14 + 0.33 = 0.49 < 1.0 \text{ OK}$$

Check lateral torsional buckling strength as per S16-09 13.8.2c:

$$\rightarrow 30/326 + 0.85(1.01)(30/77.7) + 1.43(1.08)(9/42.5)$$

$$\rightarrow 0.09 + 0.33 + 0.33 = 0.75 < 1.0 \text{ OK}$$

**PIPERACK STRUT AND BRACE MEMBER DESIGN EXAMPLE
USING CANADIAN BUILDING CODE LOAD COMBINATIONS**

Check biaxial bending of the member as per S16-09 13.8.2:

$$\rightarrow M_{fx}/M_{rx} + M_{fy}/M_{ry} < 1.0$$

$$\rightarrow 30/77.7 + 9/42.5$$

$$\rightarrow 0.39 + 0.21 = 0.5 < 1.0 \text{ OK}$$

Check deflection at midspan of the beam due to unfactored piping loads:

Unfactor the given loads by an assumed 1.25 factor to represent unfactored loads:

$$\Delta_{\max,x} = PL^3/48EI_x = (20000/1.25 \times 6000^3)/(48 \times 200000 \times 85.1 \times 10^6) = 4.2\text{mm} *$$

$$\Delta_{\max,y} = PL^3/48EI_y = (6000/1.25 \times 6000^3)/(48 \times 200000 \times 7.27 \times 10^6) = 14.9\text{mm} *$$

*deflections must be compared against acceptable deflection values as determined with pipe stress. If the calculated deflections are too large, the member size must be increased or possibly braced.

Use W310x39 for member 2

BRACE DESIGN:

Chevron style (inverted V-shape) bracing has been selected for use on the piperack. This will result in a 2:1 H:V ratio of the for the strut. This shallow angle is near the practical limit that should be chosen such that gusset plate sizes do not become too large. This style of bracing uses relatively shorter brace lengths which allow for the usage of lighter weight members.

W-shape type of bracing should not be necessary for the relatively small loads and unsupported lengths shown. Single angle type bracing is difficult to meet the slenderness requirements and secondary bending requirements. WT style bracing or 2L style bracing would be more appropriate. WT's are a single piece making installation simpler but a 2L can take advantage of double shear of the bolts. To take advantage of the compressive resistance tables in the CISC handbook, use 2L style V-Braces and apply the assumptions made therein.

Note that in some arrangements, it may be necessary to check the V-Brace with secondary bending moments that result from the eccentricity between the connection centroid and the mass centroid of the brace member.

V-Brace 1: (from CISC handbook page 4-130)

$$C_{r,2L76 \times 51 \times 6.4 \times 4\text{m long}} = 88.2\text{kN} > C_r = 36\text{kN} \text{ OK (use minimum 6mm leg thickness)}$$

V-Brace 2: (from CISC handbook page 4-130)

$$C_{r,2L76 \times 51 \times 7.9 \times 4.5\text{m long}} = 89.3\text{kN} > C_r = 85\text{kN} \text{ OK}$$

Use 2L76x51x6.4 for V-Brace 1, and use 2L76x51x7.9 for V-Brace 2



PIPE STRESS INTERFACE GUIDELINES

This attachment outlines guidelines for interface between Pipe Stress and structural engineering in order to facilitate a better understanding and communication of stress loads and analysis. The topics covered by this Attachment include:

- Overview of pipe stress
- Importance of early alignment between Pipe Stress and Structural
- Concepts related to application of reported Pipe Stress loads
- Determination of assumed loads, and structural design for these loads

PIPE STRESS OVERVIEW

Pipe Stress analysis is the process of evaluating the integrity of a given piping system. Pipe Stress objectives are to prevent failure of the piping systems, leakage at joints, and overload at connected equipment. If relative deflections between equipment connected pipe must be strictly limited then Pipe Stress should identify these locations to Structural. The stresses in the piping system are not always the governing factor for design. Pipe Stress Engineers spend the majority of their analysis time attempting to minimize the loads the piping system imposes on the restraints and mechanical equipment. In fact, Pipe Stress is often dictated by the need to minimize the thermal loads transmitted to Structural.

Pipes will expand/contract as the temperature of the pipe changes from the installed to the operating condition. Be aware that different piping materials expand/contract at different rates. For example, fiberglass and similar materials can expand at 3 or more times typical steel piping systems.

Not all lines are analyzed by Pipe Stress. Typically, the support, routing, and piping flexibility for the majority of lines are the responsibility of piping design. Pipe Stress scope is determined by line size, temperature, connections to load sensitive equipment, or other contractual specifications. To determine if a line requires Pipe Stress review, locate the project master line list. Typically the line list will indicate if a stress analysis is required. If Pipe Stress determines that additional supports are required or if the loads on a field support (provided by Piping) exceed their rated capacity, Piping Design is notified and expected to relay the requirements to Structural.

Anchors and Guides

Directional anchors and guides are used to restrain piping expansion in order to prevent overstress of the system and/or excessive loads on equipment. Anchors and guides are used in Pipe Stress analysis to establish "fixed points" in the system where movement is



PIPE STRESS INTERFACE GUIDELINES

controlled. An anchor is a mechanical interface between the piping system and an adjacent structural member.

Directional Anchors:

- Restrain axial movement of the pipe
- Allow lateral movement of the pipe
- Typical piping callout is 5DA* (*=1,2,3, or 4 depending on the line size)
- Stiffeners are often required in the supporting beam at anchor locations to prevent local buckling of the beam flange and bending of the beam web. Stiffeners do not resist torsion.

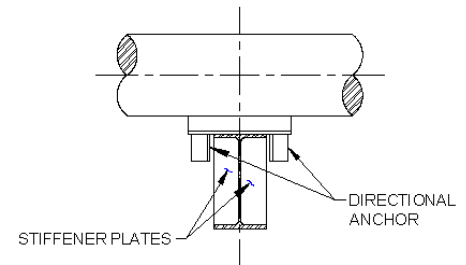
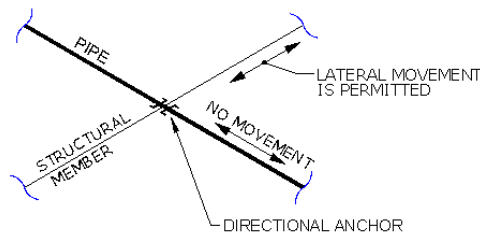


Figure 1: Directional Anchor

Guides:

- Restrain lateral movement of the pipe
- Allow axial movement of the pipe
- Typical piping callout is 5G* (*=1, 2, or 3 depending on the line size)
- Are typically provided at a standard spacing in pipe racks based on line size



PIPE STRESS INTERFACE GUIDELINES

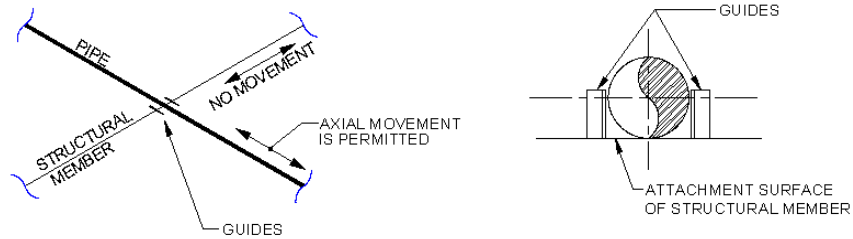


Figure 2: Guide

A directional anchor combined with a guide is commonly referred to as an “anchor”. In older designs, it was common to weld a pipe shoe to steel to establish a “full anchor” in order to restrain rotation of the pipe as well. This approach is rare in current design methodology; however it is common practice to indicate the X symbol for a full anchor on Pipe Stress Isometrics (ISOs) to represent a two directional anchor.

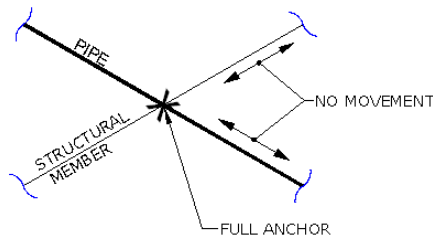


Figure 3: Full Anchor (Restrained Rotation)

Piping Layout / Design

Flexibility of a piping system is key to ensuring that anchor and guide loads are not excessive. As a pipe heats up, thermal force develops between two anchor points. See Figure 4, below where a pipe is anchored in two places. In this case, the anchor loads due to axial expansion would be maximized because there is no inherent flexibility in the piping configuration as shown. The magnitude of the anchor loads are directly proportional to the pipe expansion length (L), the difference in temperature (ΔT), and the pipe cross section (A_p).



PIPE STRESS INTERFACE GUIDELINES

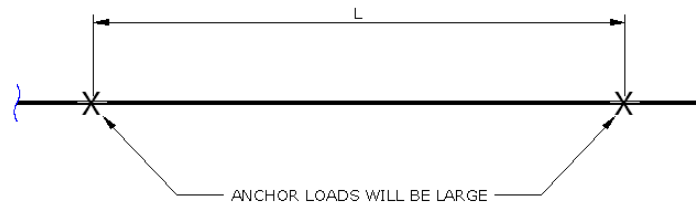


Figure 4: Example Anchor Configuration

Piping systems flex when absorbing thermal expansion. Figure 5 illustrates the deflected shape for various piping layouts due to thermal expansion.

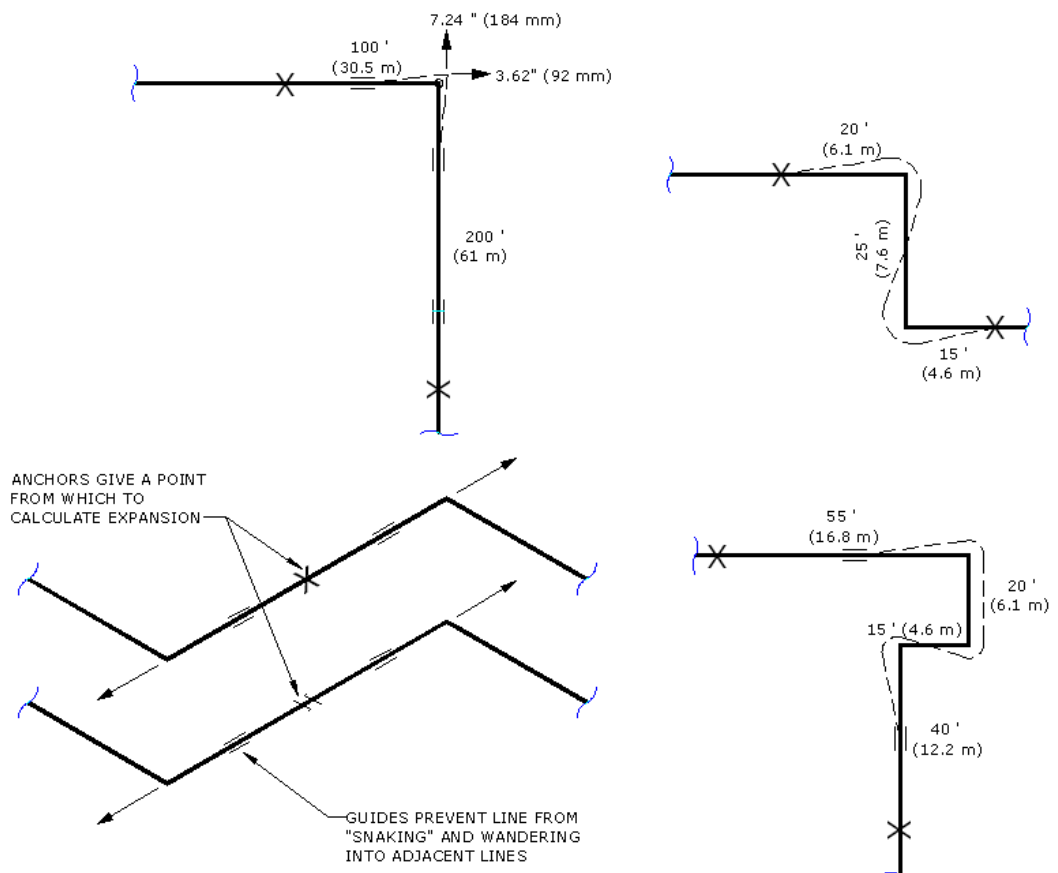


Figure 5: Piping Flex Due To Thermal Expansion



PIPE STRESS INTERFACE GUIDELINES

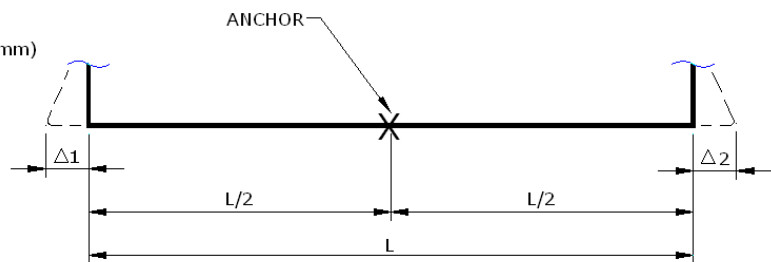
Piping systems may require loops to absorb thermal expansion and/or reduce the anchor forces. Loops add additional flexibility to the system and allow the thermal forces to be partially relieved. In order to avoid contact between pipes or contact between the pipes and structures there are a few piping general rules of thumb for the placement of loops:

- Add expansion loops in long piping runs to prevent excessive pipe movement into intersections.
- Do not allow more than 5 to 6 inches (125 to 150 mm) of pipe expansion into intersections.
- Do not allow more than 10 to 12 inches (250 to 300 mm) of pipe expansion into loops.
- Loops need to be “balanced” with approximately equal growth pushing from each side.

Figure 6 below exemplifies how adding a loop to a piping system can provide a good solution for excessive thermal growth.

Poor Solution:

$\Delta 1$ & $\Delta 2 = 8"$ (200 mm)



Better Solution:

$\Delta 3$ & $\Delta 4 = 2"$ (50 mm)

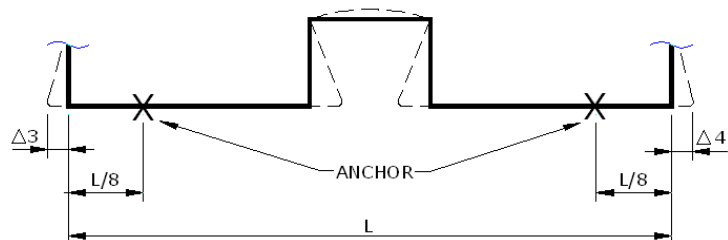


Figure 6: Adding Loops For Expansion

Expansion loops require anchors and guides to function properly. The location of anchors and guides can be optimized to minimize the thermal loads. The most critical dimension of the loop is the “H” dimension or the distance a loop extends across a pipe rack, see Figure 7.



PIPE STRESS INTERFACE GUIDELINES

In order to push the maximum expansion into a loop and maintain reasonable anchor loads, “H” must be adequate. The second most critical loop dimension is “X”, or the distance from the loop to the first guide. Pinching the guides too close to the loop generates un-necessarily large guide loads.

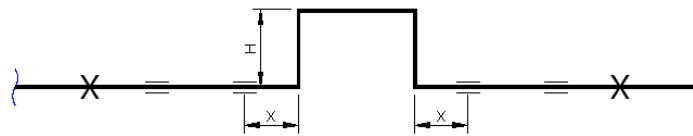


Figure 7: Loop Dimensions

The location of pipe loops should be coordinated with the location of structured pipe rack expansion joints. This may be especially important when multiple braced bays are involved.

More than one loop may be required when the forces to bend the loop are too great and the anchors cannot be economically reinforced. Figure 8 shown below illustrates this point.

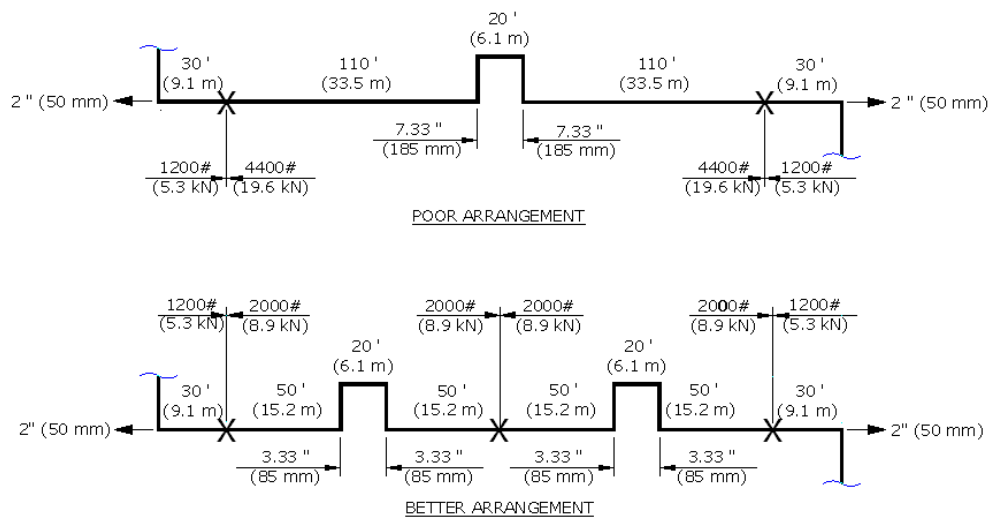


Figure 8: Use of Multiple Loops

Pipe Stress Engineers pay close attention to systems containing “load sensitive” equipment. The failure mode can be excessive nozzle loads, base plate distortion, rotor-dynamic



PIPE STRESS INTERFACE GUIDELINES

vibration, or and/or wear of the seals, bearings, or couplings. Centrifugal equipment such as pumps, turbines, compressors, and blowers can be cause for concern. Shutdown or failure of this equipment can be catastrophic and extremely expensive.

EARLY ALIGNMENT

Early alignment between Piping Design, Pipe Stress and Structural groups is a key step in making a project successful. It is critical to hold an early alignment meeting on all projects to discuss key issues and work procedures. As a minimum, the Piping Design Lead, the Lead Structural Engineer, and the Lead Stress Engineer should be present. It is strongly recommended that area leads and other key players participate as well. Objectives of early alignment meetings are to:

- Identify structural priorities and need dates.
- Identify Potential schedule conflicts.
- Develop a stress priority log based on steel and foundation issue dates. Review this priority log with Engineering management. Be clear and indicate on the schedule when critical Pipe Stress will be received as milestones built into the schedule.
- Break the scope of work into logical sections based on the nature of the structural work.
- Identify which load cases need to be reported to Structural. There needs to be an understanding of the loads that are given and how structural should apply them. For example, do the thermal loads provided include friction?
- Discuss that correct signage for loads is important and identify if the loads are reversible.
- Emphasize that Structural would like to see actual loads instead of only the maximum or enveloped loads. For example, the empty dead load reported may be greater than the operating dead load when thermal effects are included.
- Establish a standard coordinate system.
- Establish structural "cutoff loads" to determine what loads are required for submittal to the Structural group.
- Define the site specific data for wind, seismic, deflection, settlement, etc.
- Clearly define what loading "bumps" [Design Margins] (increase factors that are not related to load factors) will be incorporated during each phase of the design. This should be addressed in the Design Margin Table as part of the Cost



PIPE STRESS INTERFACE GUIDELINES

Competitive Engineering and Design (CCE&D) effort on the project. Refer to 000.200.1400 and 000.200.1401 for CCE&D framework and design margin tables, respectively.

- Clearly define whether the provided pipe stress loads are allowable stress loads or factored Strength level loads.
- Discuss the use of stiffener plates at anchor locations. Clarify which group will provide them. It is best for Structural to provide the stiffeners if possible. If schedule does not allow Structural to include stiffeners in steel drawings, piping will need to reflect these on their ISOs. Be aware that field installation of these stiffeners can be very expensive and field installation on fireproofed steel may be very difficult. Stiffeners should only be provided when calculations indicate they are necessary. Discuss whether stiffeners are required on solid boxed concrete fireproofed structural members or anchors can bear directly on the concrete.
- Define any special design issues such as implementation of large modules or tall slender structures.
- Establish a Miscellaneous Pipe Support (MPS) procedure.
- Identify areas where there is potential for differential vertical settlement between adjacent foundations and how these settlements may affect the pipe stress analysis.

An example of some project specific interface guidelines is shown below:

Example: Project Specific Pipe Stress Criteria (using ASCE 7-05)

(NOTE: this criteria will vary from one project to another)

- *Magnitude of loads reported to Structural*
 - *Report loads > 1 kip(4kN) for elevated vertical loads*
 - *Report loads > 0.5 kip(2kN) for horizontal loads*
- *Load cases for reporting*
 - *Dead loads (empty Dead Wt. and operating Dead Wt. only)*
 - *Max. Oper. (operating Dead Wt. + Thermal)*
 - *Friction (direction sensitive)*
 - *Hydrotest (Report loads higher than max. oper. Loads)*
 - *Seismic (not including dead, thermal, etc.)*
 - *Wind (not including dead, thermal, etc)*
 - *Occasional (OCC). (Special case when applicable, relief valve, slug force, etc.)*
- *Coordinate System*
 - *X-axis = + to East*
 - *Y-axis = + Vertical*
 - *Z-axis = + to South*



PIPE STRESS INTERFACE GUIDELINES

- *Seismic Load Criteria*
 - $F_p = 0.17$
 - *Reported forces should be strength level (do not multiply by 0.7)*

LOADS

The Pipe Stress Engineer can report any of the load types listed below. The design Engineer should check to determine if a project structural bulletin has been issued to address the Pipe Stress reported loads.

- Empty Dead
- Operating Dead
- Hydrotest
- Wind
- Seismic
- Thermal
- Friction
- Special (Relief Valve Reactions, Surge, Slug Flow, etc)

Pipe Stress may utilize several different methods of load reporting to Structural. This will vary on a project to project basis. Typically there will be a stress ISO with nodes and an associated load sheet indicating the design loads.

Application of Reported Pipe Stress Loads

It is important to understand and apply the loads provided by Pipe Stress appropriately to avoid over design of the structure.

Dead Loads

- Typically structures should be analyzed with dead loads calculated by the Structural Engineer and not the provided Pipe Stress dead loads unless:
 - Uplift is a significant factor (i.e. loads can redistribute or hold down clamps may be provided). [see Note 1]
 - There are piping components that cannot be adequately accounted for such as large flanges and valves.
 - Thermal loads significantly redistribute the dead load, see Figure 9 below. [see Note 1]



PIPE STRESS INTERFACE GUIDELINES

[Note 1: It should be clarified during the alignment meeting that either reported “Dead Loads” & “Thermal Loads” should be added together or that reported “Thermal Loads” include dead loads.]

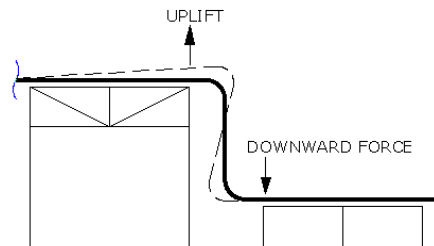


Figure 9: Dead Load Redistribution Due to Thermal

Thermal Loads

- Thermal loads should be applied with the magnitude and specific direction that is indicated on the sketches provided by Pipe Stress. Be aware that thermal loads can occur in multiple directions simultaneously. It is important to apply the loads in the direction given to fully account for any tension or compression in the structure that may result from the load opposition.
- The Structural Engineer should make sure that a reasonable provision has been made for unbalanced friction loads. The Structural Engineer should check for a project bulletin or ask Pipe Stress to determine if the thermal loads provided include unbalanced friction. Unbalanced friction occurs when the friction force at supports on either side of an anchor are not balanced, see Figure 10. Pipe Stress may provide the actual unbalanced friction based on the layout of the supports, an estimate of the unbalanced load based on a certain assumed number of unbalanced supports, or unbalanced friction forces may not be included in the provided loads. If Pipe Stress provides unbalanced friction loads based on an estimated number of supports, the loads applied in the structural model may be reduced or eliminated depending on the actual layout and if the anchors are balanced. If unbalanced friction forces are not reported by Pipe Stress, the Structural Engineer should include estimated loads in the structural design.



PIPE STRESS INTERFACE GUIDELINES

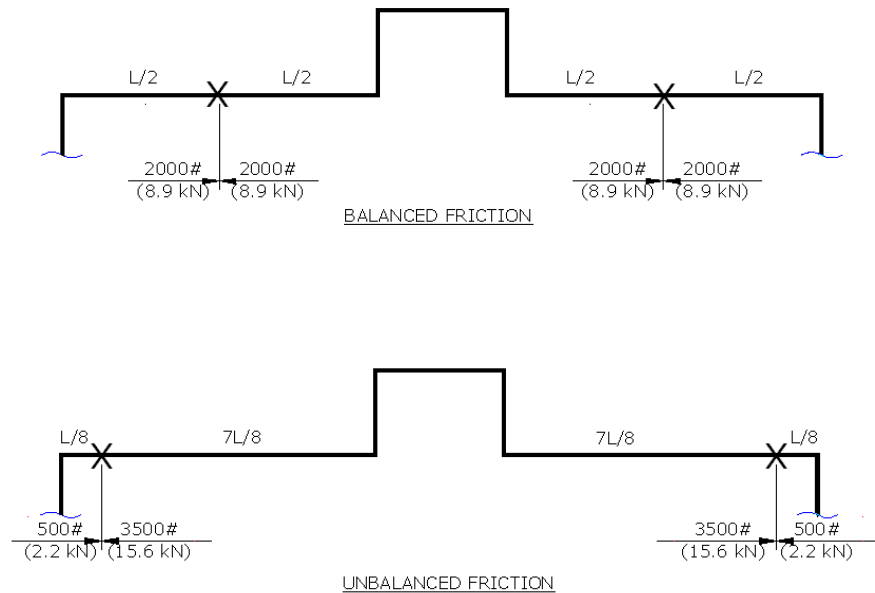


Figure 10: Unbalanced Friction

- Thermal loads may not need to be carried to all members and foundations. Consult the Lead Engineer for guidance.
- If the friction load for a large pipe is given by Pipe Stress at an anchor point located in an independent action portion of the rack, the uniform distributed friction load of this particular line need not be considered at other supports located within this independent portion of the rack. See Figure 11 below for an illustration of this load application.



PIPE STRESS INTERFACE GUIDELINES

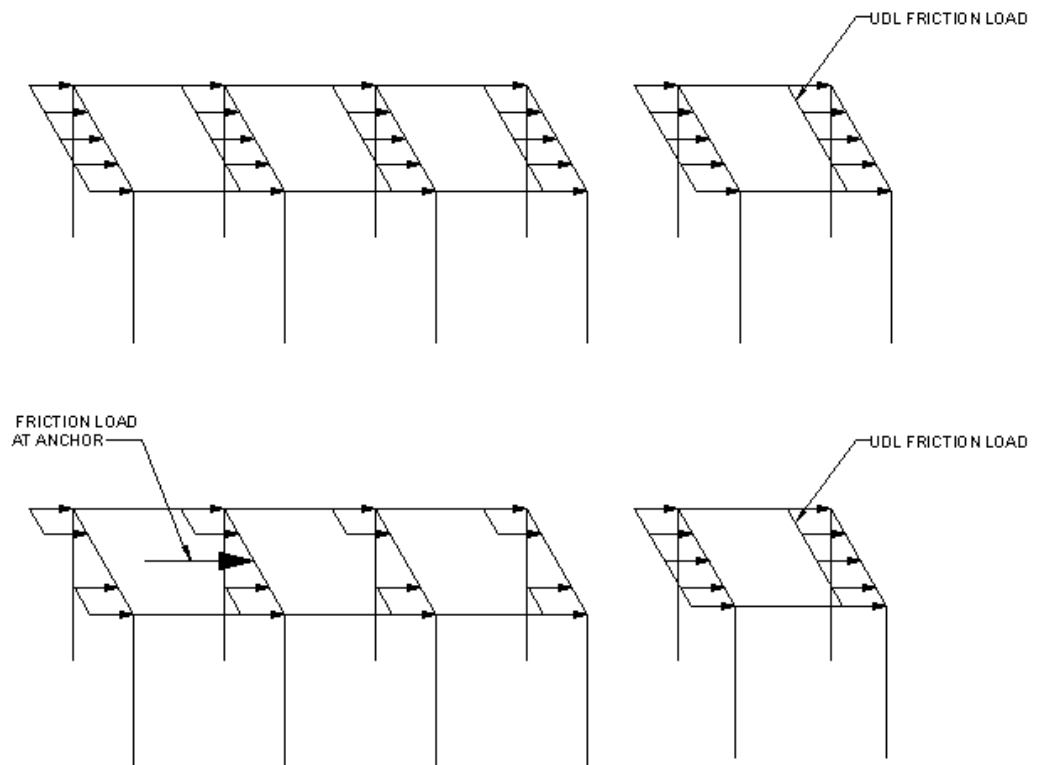


Figure 11: Application of Unbalanced Friction Load in a Rack

Wind Loads

- Wind loads provided by Pipe Stress should only be used for design of local members and their connections. For design of the overall structure, the wind loads calculated by Structural should be used.

Seismic Loads

- Seismic loads provided by Pipe Stress should only be used for design of local members and their connections. For design of the overall structure, the seismic loads calculated by Structural should be used.



PIPE STRESS INTERFACE GUIDELINES

- A load path must be provided between the anchor and the structure lateral force resisting system.
- The Structural Engineer should be aware of the basis of the seismic coefficients used by the Pipe Stress Group. For example, when using ASCE 7, the seismic loads reported from Pipe Stress are typically based on the nonstructural component seismic design force, F_p . This design force is different than the seismic loads calculated for the structure which are based on the seismic response coefficient, C_s . The nonstructural component force should be used for the local supporting members and connections only.
- Be aware of the basis of the seismic code provisions being used in order to avoid unnecessary compounding of factors on loads and overly conservative results.

FOR EXAMPLE: [ASCE 7-2005] The use of the F_p to derive seismic coefficients used by Pipe Stress may be conservative in some cases. See below for F_p equations according to ASCE 7-05 for cases where the weight of the piping system is less than 25% of the total effective weight of the structure. The minimum equation typically governs. Often $z = h$ assumption is made in order to have one seismic coefficient for all cases. To simplify Pipe Stress efforts, often a conservative importance factor may be used for all pipes. Components containing hazardous materials the importance factor, I_p , is equal to 1.5 per ASCE 7-05, 13.1.3. However, if the pipe does not contain hazardous materials, I_p shall be equal to 1.0. Be careful to ensure that for multiple pipes supported on a structure, the importance factor used may need to be higher for all the anchor / guide loads to ensure that the hazardous lines are protected.

$$F_p = \frac{0.4a_p S_{DS} W_p}{\left(\frac{R_p}{I_p}\right)} \left(1 + 2\frac{z}{h}\right) \quad (13.3-1)$$

F_p is not required to be taken as greater than

$$F_p = 1.6 S_{DS} I_p W_p \quad (13.3-2)$$

and F_p shall not be taken as less than

$$F_p = 0.3 S_{DS} I_p W_p \quad (13.3-3)$$

Assumed Pipe Stress Loads

Often the actual pipe stress loading is not available during the initial design of pipe supports. For certain pipes, it may be necessary to assume Pipe Stress loads in order to avoid field modifications to pipe supports. The Lead Engineer should be consulted to determine a general design philosophy for assumed loads and/or identify any project guidelines or procedures that may already be established.



PIPE STRESS INTERFACE GUIDELINES

Assumed pipe stress loads should be made as part of the CCE&D effort at the beginning of the project and a project bulletin should be issued to address assumed pipe stress loading when no information is available. Reference should be made to 000.200.1400 practice.

When assuming Pipe Stress loads, it is best to start with the project master line list and use some good engineering judgment. Some general concepts to remember are:

- High temperature lines = high thermal anchor loads.
- Large, Heavy Lines = High Seismic Anchor Loads. It is always a good idea to obtain process input for pipe content specific gravities before estimating the seismic anchor load. Watch out for piping contents with specific gravities > 1.0 (heavier than water).
- High pressure lines have thicker walls, are less flexible, and will tend to yield higher thermal loads

If it is determined that Pipe Stress loading may be critical for a project and it is necessary to incorporate assumed loads in design for all or the majority of pipes, it is suggested to develop a piping load table based on the line list. This piping load table can be established on a project basis by using the line list spreadsheet (obtained from piping) and incorporating new columns for assumed thermal and seismic anchor and guide loads. The assumed thermal loads should be reviewed with Pipe Stress to determine what is reasonable for design. Seismic anchor and guide loads are just a function of the piping weight and an assumed anchor spacing. These can be easily estimated by the Structural Engineer. See Figure 12 below for an example Piping Load Table.

service	line no.	size	line class	sched	We (plf)	Wt (plf)	Wo (plf)	specific gravity	phase	P&ID	Assumed Thermal Anchor Load (k)	Assumed Thermal Guide Load (k)	design T	Assumed Seismic Anchor Load (k)	Assumed Seismic Guide Load (k)
PL	1003	4	BAAA2	XS	15	20	20	1	L	43318	1.5	1	250	4.0	1.0
PL	1005	12	BAAA2	XS	83	130	130	1	L	59718	4.8	2.4	400	8.0	3.0
HPS	1008	16	PAAA1	80	158	227	165	0.1	V	59713	8.5	5.5	800	6.6	6.6

Figure 12: Piping Load Table (with assumed loads)

Once some generic assumed loads are identified, some judgment can be used to assume the locations for application of these loads. Discuss with Pipe Stress to determine if preliminary anchor and guide locations are available and/or identified in the 3D model. If these locations are not available, review the plot plan to identify a general layout for loops and anchors. Pipe Stress should be able to provide some guidance for where the anchors and loops will be located at a cursory review of the piping layout on the plot plan. Be aware that terminal anchors will yield higher thermal loads. Terminal anchors are anchors that are located



PIPE STRESS INTERFACE GUIDELINES

nearest to a piping change in direction. The loads at terminal anchors are higher since the forces due to the pipe expansion are not balanced on either side (see Figure 13 below).

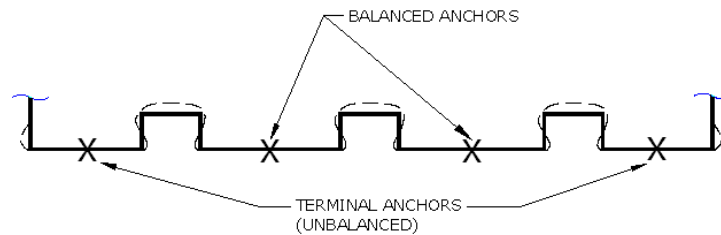


Figure 13: Terminal Anchors

Once approximate anchor locations have been established, review the structural layout to determine if “strong points” are present in the near vicinity of these locations. A “strong point” indicates a place in the structure that by configuration would be more able to resist lateral anchor and guide loads. This is typically a braced anchor bay in a pipe rack or a braced location at or near a column line in a structure.

The assumed anchor and guide loads and locations shall be validated against the final actual stress information upon its availability. Validation may be done by comparing the actual stress load values with the assumed load values. Re-analysis is recommended where actual stress loads are more critical than the assumed loads, or cost optimization can be achieved.

Occasionally it is necessary to assume Pipe Stress loads for pipes where no formal Pipe Stress analysis is required. This means that the lines are typically low temperature/pressure and are smaller in diameter. The thermal loads at the anchors and guides will be small; however for some lines, it is important to consider the seismic anchor loads for local beam design (particularly in existing areas). Typically ambient temperature water lines 8 inches (200 mm) in diameter or less are not analyzed by Pipe Stress. However, the anchor spacing may be large enough that the seismic anchor loads are important to consider for design. The seismic anchor load is relatively easy to estimate by calculating a tributary length of pipe to consider as weight acting at an anchor in conjunction with the F_p coefficient used by piping for component seismic loads. Use judgment and consult the Lead Structural Engineer to determine when seismic anchor loads need to be estimated for pipes with no stress analysis.

DESIGNING FOR PIPE STRESS LOADS

There are a few things to keep in mind when designing for Pipe Stress loads. Good initial set up of the structural model can make implementing stress loads much easier. When Pipe Stress loads are a significant factor in driving the structural design, it may be necessary to refine calculations or implement changes to the piping and/or structural layout to achieve a more cost effective configuration.



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Setting up Structural Analysis Software model

It is good practice to orient the structural analysis software model in the direction similar to the stress load output. Typically Z points South. It is also recommended to input Pipe Stress loads as joint loads in the structural analysis software model by creating joints named directly after the ISO number or stress node number. This will make it easier to input stress load revisions, determine displacement at support locations, and facilitate checking.

Pipe Stress loads should typically be applied in the direction given. Do not apply all the loads in the same direction unless they are reported that way. Watch out for cases where the resultant anchor/guide force on a structure is in the negative direction. If the load combinations utilized reverse the Pipe Stress loads, it may be necessary to reverse the applied direction of the Pipe Stress loads since other transient loads are applied in the positive direction by default.

When Loads are Problematic

In the event that the reported Pipe Stress loads are problematic for the structure, there are a few strategies that may be implemented to determine a solution.

- Ask Pipe Stress to add loops and anchor more often. The trade off for this solution is that additional supports will be required to support the new loops.
- Ask Pipe Stress to stagger the anchors to avoid multiple supports on one beam.
- If structural modifications cannot be avoided, it may be better to anchor all pipes on one beam or braced bay to limit the structural upgrades to one location.
- Ask Pipe Stress to move anchors closer to columns or better support locations such as braced bays.
- Provided a spring stiffness representative of the structure flexibility for Pipe Stress to model as a part of their analysis to simulate the structure flexibility and reduce the stress loads. Caution should be used when providing spring stiffnesses to Pipe Stress for the structural support locations. This can become an iterative process. For example, the Structural Engineer may provide a spring stiffness to Pipe Stress which will then reduce the loads that Pipe Stress provides back to Structural. The Structural Engineer may then elect to reduce member sizes, inherently increasing the structural flexibility. The Structural Engineer will then need to report the new spring stiffness back to Pipe Stress so that the piping system can be re-evaluated for the new flexibility.
- To determine the spring stiffness to be provided to Pipe Stress, follow the steps outlined below.



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- STEP 1: Add a node in the structural model at the location where spring stiffness is desired.*
- STEP 2: Apply an appropriate unit load (i.e. 1 kip, 10 kips, etc. to achieve a reasonable deflection) in the direction that a stiffness is desired.*
- STEP 3: Create and run a load combination for just the applied unit load (no structure self weight or dead load).*
- STEP 4: Divide the applied unit load by the joint deflection to obtain the spring stiffness.*

Any time Structural requests changes in piping layout, there needs to be a dialogue between piping design, Pipe Stress, and structural to determine what solution best meets the needs of the project. Discuss with the Lead Structural Engineer what design philosophy should be used for particular projects and structures.

Structural Support Deflections

Pipe Stress analysis involves incorporating fixed boundary conditions for anchors and guides in a piping design model. The Pipe Stress analysis assumes there is no deflection at these locations. Deflections at structural supports are unavoidable. In most piping systems, some support lateral deflection is acceptable for the piping system. The specific criteria for deflection will be as specified in project requirements as determined by the Lead Engineer. Generally the local deflection at the anchor points shall be restricted to ½ inch (12 mm). Deflection should be calculated and reviewed with the Pipe Stress Engineer if:

- The Pipe Stress engineer indicates that the deflection of a particular anchor is critical (may occur near load sensitive equipment)
- Multiple anchors are located on the same support
- There is an absence of other lines to provide restraint

If the deflection requirements for Pipe Stress cannot be met, it is generally adequate to provide plan bracing to the anchor point. The anchor location can then be considered braced and deflection at the anchor need not be reviewed for Pipe Stress requirements.

Excessive vertical deflection should also be limited to avoid sagging of the structure, pipe lifting off or the supports, and load redistribution.

In special cases, such as very tall flexible structures, it may be necessary to provide structural support spring stiffness to piping or a combined piping/structural steel model may need to be analyzed so that Pipe Stress can more accurately account for the structural deflections in their analysis.



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Deflection due to settlement should also be addressed for structural analysis and/or piping design. For the Pipe Stress/structural interface, usually only long term settlement due to sustained loads is a potential issue. Within operating units, differential settlement should typically not be an issue if good design procedures are followed. OSBL (Outside Battery Limits) installations, such as tanks and large basins, can provide settlement challenges that must be addressed. Interface between Pipe Stress and Structural is required to manage the effects of relative vertical settlement between adjacent foundations such as pumps adjacent to pipe supports.

Displacements Between Structures

It is important to consider seismic relative displacements between structures for the design of the structural and piping systems. In a seismic event, two adjacent independent structures can deflect in opposing directions developing additional forces in the piping systems supported between the two structures. If there is not enough piping flexibility between the two structures, the stresses in the pipe and/or the lateral loads at the supports may be problematic. Therefore, it is always a good idea to build some piping flexibility into the configuration between two adjacent structures (especially if the structures are tall and flexible where seismic drifts may be large). Piping flexibility can be achieved by avoiding anchoring pipes on directly adjacent structures or adding loops. In special cases, flexible connections can be built into the pipe, although this can be an expensive alternative. See Figure 14 below where this concept is illustrated. In cases where seismic displacements between structures are a concern, there needs to be a joint discussion between the Structural and Pipe Stress groups to determine the optimal solution.



PIPE STRESS INTERFACE GUIDELINES

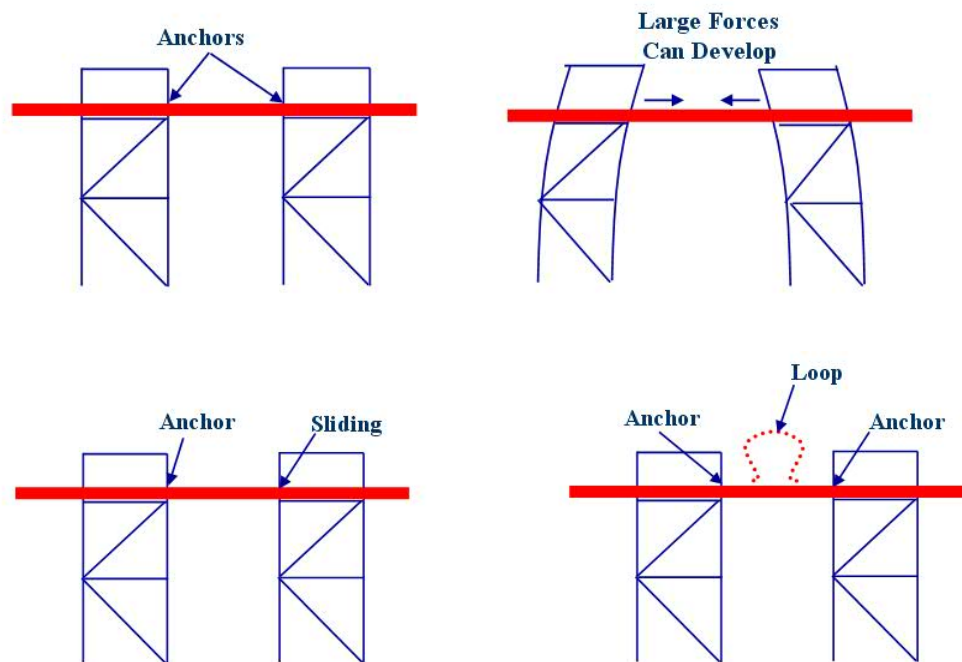


Figure 14: Seismic Displacement Between Structures – Piping Flexibility

The seismic relative displacement to consider for analysis shall be determined by adding the deflection magnitudes determined by elastic analysis for each independent structure (per ASCE 7-05 Section 13.3.2.2). Deflections determined by elastic analysis do not include the deflection amplification factor or importance factor as defined in ASCE 7-05 Section 12.8.6.

The relative stiffness between the two structures or vessels is also important to consider. If one structure is significantly more flexible than the other, it will be driven by the more rigid structure and relative motion may not be an issue. Structures with similar stiffness may need more consideration.

Designing the pipe and structure for independent motion between two structures (lateral displacement within the same plane) may be conservative for direct 2D motion. It may even be too conservative to stiffen the structures and design the pipe for this relative motion. This approach neglects the contribution of the pipe to link the two structures and resist relative displacement. It may be more cost effective to consider large pipes as structural links and do a more detailed analysis of the linked structure as a whole. Two independent RISA-3D models can be appended together with the rigid pipes modeled between the two structures. Judgment should be used to determine which pipes are rigid enough to qualify as links. This will require more engineering and analysis time. The lead should be consulted for direction.



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Also, be aware of torsional effects for piping located between two structures. Torsional loads may cause higher nozzle loads than expected. Pipe Stress needs to be made aware of these issues to protect the equipment nozzles. See Figure 15 below demonstrating this effect.

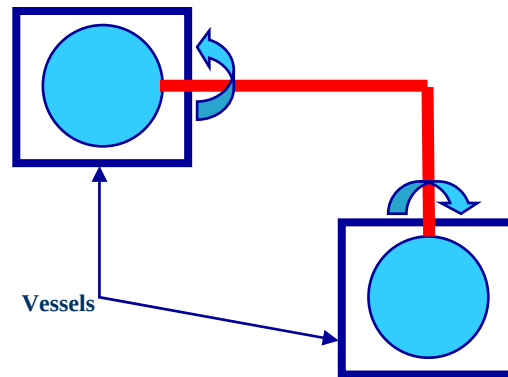


Figure 15: Example: Torsion on Adjacent Structures

Cost Effective Design

Pipe Stress and Structural Engineers must develop a joint design philosophy in order to produce the most cost effective design. Strategies that may be employed to provide a more cost effective design when Pipe Stress loads are not available are:

- Design the structure without Pipe Stress loads and create a standard horizontal bracing detail to add when loads are available. (Include assumed thermal loads or contingency for foundation design). Be aware of schedule constraints. Field installation of supports can be very costly and time consuming.
- Layout structures with strong locations for anchors. Communication / discuss these locations with Pipe Stress.
- Also, avoid double dipping assumed or provided Pipe Stress loads for dead, wind, and seismic with the loads calculated and applied by Structural.
- Align with pipe stress that large pipes should not be supported on intermediate beams. Intermediate beams are only intended to support small pipes (pipes that can not span between main bents)
- Cost effectiveness can only be achieved if Pipe Stress loads are provided to Structural in a timely manner and Structural is not forced to do their work before the pipe stress information is available. If this can not be achieved then schedules and budgets should reflect this work process and Project Management should acknowledge the impacts.